5 Time Response Analysis of Control Systems

Objective

After completing this chapter,the user will be able to

- Determine the response of a system whose transfer function is given for any input.
- Determine the transient response for systems of a different order.
- Determine various time response specifications for second-order under damped systems.
- Understand the concept of error calculation.
- Understand sensitivity concepts.

Introduction

- The input to a control system is not known ahead of time, and it is random in nature. So we may not be able to express it analytically in terms of mathematical functions or waveforms.
- If the objective of the control system is to have the output variable follow the input signal as closely as possible, it is necessary to compare input and output as functions of time.
- So, we need criteria for comparison of various control systems for which we apply some standard test signals and compare the responses of control systems to these test inputs.
- The commonly used test signals are step input, ramp input, parabolic input, impulse input, and sinusoidal input.

Time response:

The output or response of a system, which is varying w.r.t. time is called time response. The time response c(t) of a control system is usually divided into two parts.

1. Transient response
2. Steady-state response

$$c(t) = c_t(t) + c_{ss}(t)$$

$c_t(t)$ = Transient Response

$c_{ss}(t)$ = Steady-state Response

In the control system, the steady-state response is simply the fixed response when time reaches, infinity or the part of the time response that remains even after the transient dies out.

Transient response is defined as part of the response that goes to zero as time becomes very large.

$$\lim_{t\to\infty} c_t(t) = 0$$

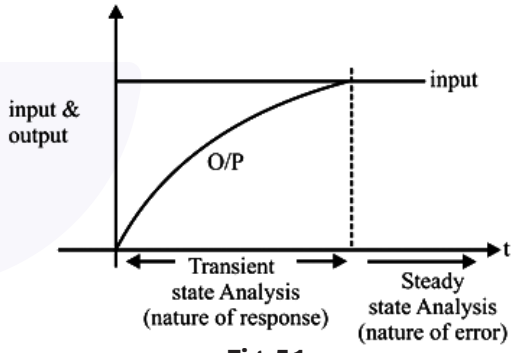
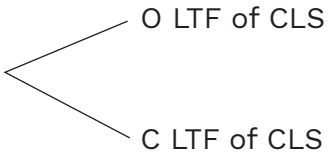


Fig. 5.1

Time response:

1. O.L.S.
2. C.L.S



3. Order of T.F.

4. Type of CLS $\xrightarrow{\text{decided by}}$ OLTF of CLS

First-order system:

1. First-order open-loop system (O.L.S):

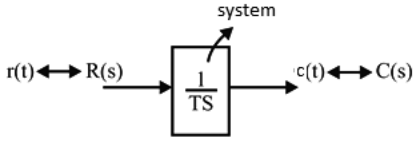


Fig. 5.2

$$T(s) = \frac{C(s)}{R(s)} = \frac{1}{Ts}$$

a) Calculate the impulse response of the above system

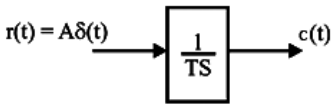


Fig. 5.3

Impulse Response

$$r(t) = A\delta(t) \leftrightarrow R(s) = A$$

$$T(s) = \frac{1}{Ts}$$

$$C(s) = R(s)T(s) = \frac{A}{Ts}$$

$$c(t) = \frac{A}{T} \text{ or } \frac{A}{T}u(t)$$

b) Calculate the output for the input

$$r(t) = A\delta(t-\tau)$$

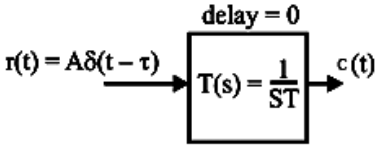


Fig. 5.4

$$C(s) = R(s)T(s)$$

$$C(s) = \frac{Ae^{-s\tau}}{sT}$$

$$c(t) = \frac{A}{T}u(t-\tau)$$

c) Calculate output ifthe system offers a delay of τ sec.

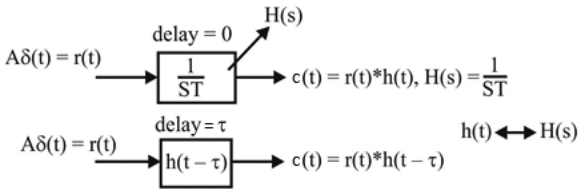


Fig. 5.5

$$C(s) = R(s)e^{-s\tau}H(s) = \frac{Ae^{-s\tau}}{ST}$$

$$c(t) = \frac{A}{T}u(t-\tau)$$

Important:

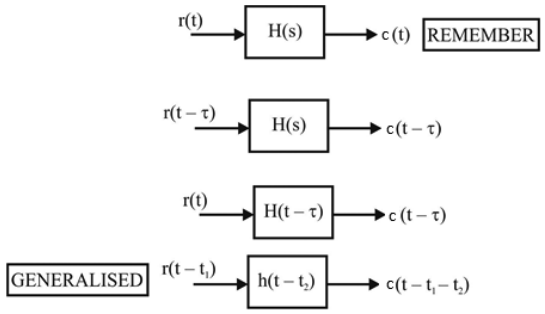


Fig. 5.6

d) Calculate step response of the above system

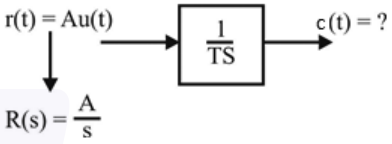


Fig. 5.7

$$C(s) = R(s)T(s) = \frac{A}{s} \times \frac{1}{TS}$$

$$C(s) = \frac{A/T}{S^2}$$

$$c(t) = \frac{A}{T}tu(t)$$

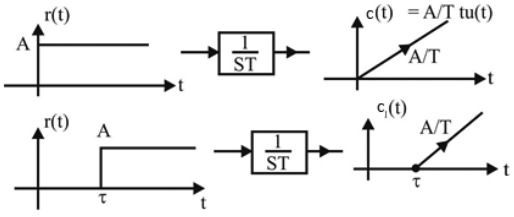


Fig. 5.8

$$c_1(t) = \frac{A}{T}(t-\tau)u(t-\tau)$$

2. First-order closed-loop system (C.L.S): [Unity negative F/b]:

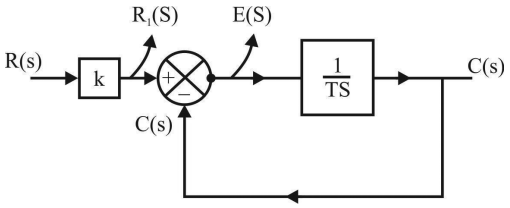


Fig. 5.9

$$T_1(s) = \frac{C(s)}{R_1(s)} = \frac{\frac{1}{TS}}{1+\frac{1}{TS}} = \frac{\frac{1}{T}}{S+\frac{1}{T}}$$

$$T(s) = \frac{C(s)}{R(s)} = \frac{\frac{K}{T}}{S+\frac{1}{T}}$$

$$\frac{C(s)}{R(s)} = T(s) = \frac{K/T}{S+\frac{1}{T}}$$

Pole-zero diagram

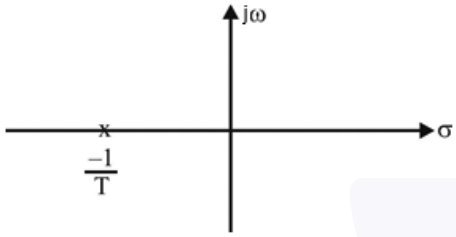


Fig. 5.10

a) Calculate the impulse response of the above system

$$T(s) = \frac{K/T}{\left(S+\frac{1}{T}\right)}$$

$$r(t) = A\delta(t) \leftrightarrow R(s) = A$$

$$C(s) = R(s)T(s) = \frac{\frac{AK}{T}}{\left(S+\frac{1}{T}\right)}$$

$$c(t) = \frac{AK}{T}e^{-t/T}u(t)$$

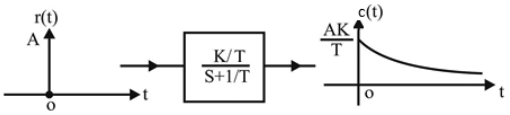


Fig. 5.11

$$C(t) = \frac{A}{T}Ke^{\frac{-t}{\tau}}$$

$$\lim_{t\to\infty} c(t) = 0 \rightarrow \textit{steady state value}$$

of impulse response $= 0$

b) Calculate step response of the above system



Fig. 5.12

$$C(s) = T(s)R(s)$$

$$C(s) = \frac{\frac{AK}{T}}{S\left(S+\frac{1}{T}\right)} = KA\left[\frac{1}{S} - \frac{1}{S+\frac{1}{T}}\right]$$

$$c(t) = AK\left(1-e^{-t/T}\right)u(t)$$

Important:

$$c(t) = AK\left(1-e^{-t/T}\right)$$

$$\left\{\frac{dc(t)}{dt}\right\}_{t=0} = \frac{AK}{T} \rightarrow \textit{slope of step response at } t = 0$$

Note: "Slope of step response represents the impulse response"

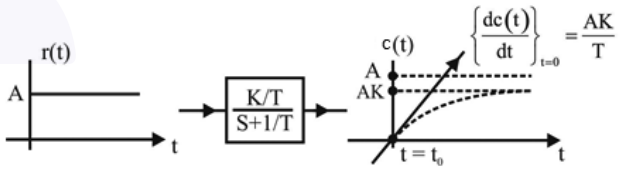


Fig. 5.13

1. If the input is r (t) = Au(t) → Then the output approaches AK
2. In the input of summer is considered $r_1(t) = AKu(t) \rightarrow$ Then output approaches AK, which is $r_1(t)$.

$$T(s) = \frac{\frac{K}{T}}{\left(S+\frac{1}{T}\right)} \xrightarrow{s=0} K \rightarrow dc\ gain$$

Note: "In any step response, output approaches or settles at input X D.C gain"

Important Terms

$$c(t) = \overbrace{AK}^{\text{constant}} - \underbrace{AKe^{\frac{-t}{T}}}_{\text{Transient Response}} \quad \text{decaying exponential dependency}$$

(i) $\lim_{t\to\infty} c(t) = \underbrace{AK}_{\text{steady state response}}$

1. Transient response:

It is the part of the total step response which tends to 0 when a large time frame is considered.

a) It tends to 0 as $t \to \infty$

b) To calculate transient response from DE,

a. $IC \neq 0$, input = 0

c) It is also known as zero input response

2. Steady-state response:

It is the part of the total step response which remains after the transient dies out.

a) To calculate the steady-state response from DE: I.C. = 0 and input $\neq 0$

b) It is also known as a zero-state response.

a. $c(t) = c_{tr}(t) + c_{ss}(t)$

c) $\lim_{t\to\infty} c(t) = c_{ss}(t)$

Example: The step response of 1storder C.L.S. is

$$c(t) = 2\left(1 - e^{-t/3}\right)$$

i. Steady-state response = 2

Transient response = $-2e^{-t/3}$

ii. $\lim_{t\to\infty} c(t) = c_{ss}(t) + c_{ss}(t)$

3. Settling time:

a) Error band

Step response

$$c(t) = A\left(1 - e^{-t/T}\right)$$

K = 1

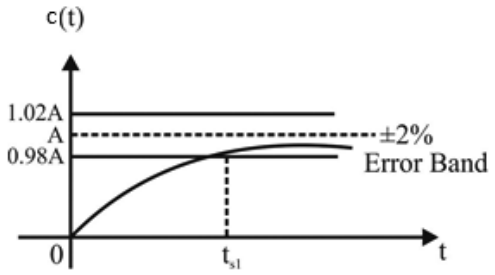


Fig. 5.14

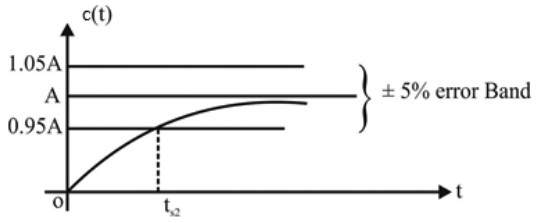


Fig. 5.15

$t_{s_1} > t_{s_2}$

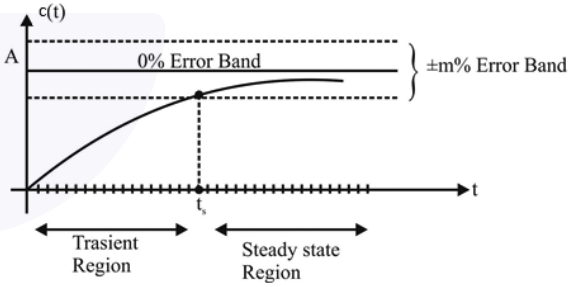


Fig. 5.16

Important Points

1. Settling time is the time required for the output to settle in a predefined error band.
2. Error Band is ± m% of input x D.C gain.
3. t_s = f(error band)—Function of error band
4. Asan error band $\uparrow \to t_s \downarrow$
5. For 0% error band $\to t_s \to \infty$
6. Upto finite settling time, transient region almost dies down.
7. After finite settling time, the steady-state region starts.

1. Calculate the time required for the step response of 1st order C.L.S. to reach 98% of its final value (steady-state value).

Solution: $c(t) = A\left(1 - e^{-t/T}\right) \quad K = 1$

1. Final value = $\lim_{t\to\infty} c(t) = A$

2. 98% of final value = 0.98A
3. At $t = t_1 \quad c(t_1) = 0.98A$

$$0.98A = A\left(1 - e^{-t_1/T}\right)$$

$t_1 = 4T \rightarrow$ Settling time for 2% error band.

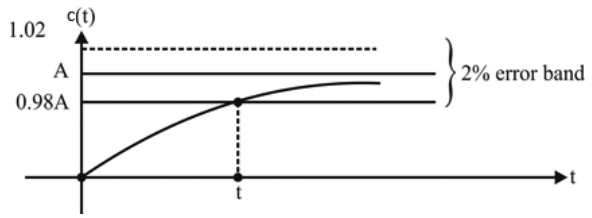


Fig. 5.17

Note: For $T(s) = \dfrac{\frac{1}{T}}{S + \frac{1}{T}}$

- Negative unity feedback
- Step response

Error Band	t_s
±2%	4T
±5%	3T

4. Error signal:

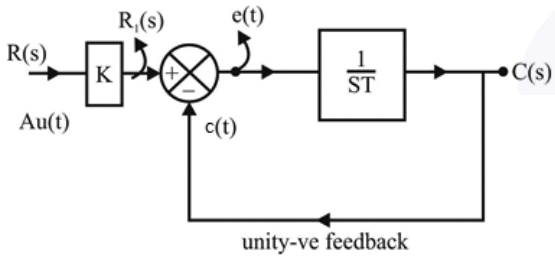


Fig. 5.18

e(t) : Error signal as long as the feedback is unity –ve.

i. $e(t) = r_1(t) - c(t)$

$$e(t) = AKu(t) - AK\left(1 - e^{\frac{-t}{T}}\right)u(t)$$

$$e(t) = AKe^{\frac{-t}{T}}u(t)$$

Steady-state error:

Final value of the error signal

$$e_{ss} = e(\infty) = \lim_{t\to\infty} e(t) = 0.$$

ii. Let $e_1(t) = r(t) - c(t)$ calculate steady state error where $e_1(t)$ *is* error signal.

$$e_1(t) = Au(t) - Ak\left(1 - e^{-t/T}\right)u(t)$$

$$e_1(t) = \left(A - Ak + Ak_e^{-t/T}\right)u(t)$$

$$(e_1)_{ss} = A(1-k)$$

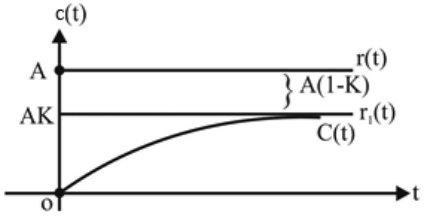


Fig. 5.19

Standard first-order circuit:

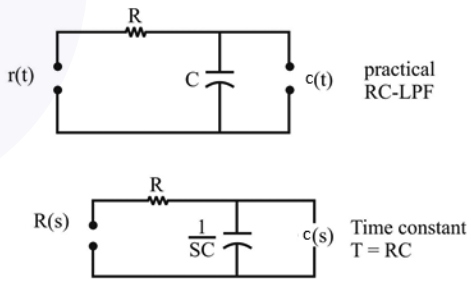


Fig. 5.20

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

Pole–zero plot:

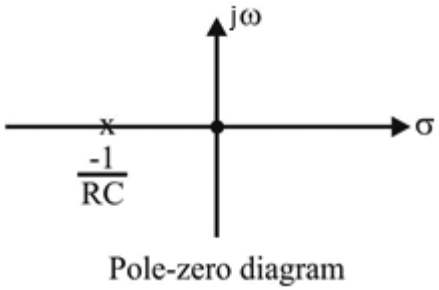


Fig. 5.21

Important: Time constant

$$= \frac{1}{|\text{Real part of dominant pole}|}$$

$$T = \frac{1}{\left|\frac{-1}{RC}\right|} = RC$$

Ex-

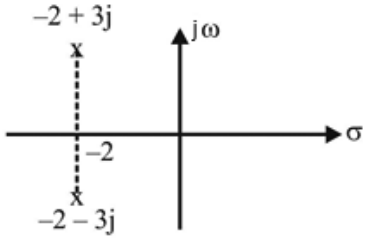


Fig. 5.22

$$T = \frac{1}{|-2|} = \frac{1}{2} = 0.5\,\text{sec}$$

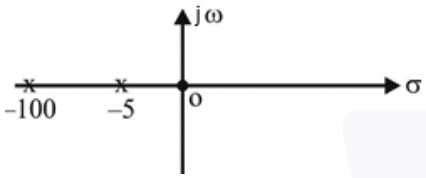


Fig. 5.23

$$T = \frac{1}{|-5|} = \frac{1}{5}\,\text{sec}$$

3-dB bandwidth of RC low pass filter:

$$H(s) = \frac{C(s)}{R(s)} = \frac{\frac{1}{RC}}{s + \frac{1}{RC}} \leftrightarrow h(t) = \frac{e^{-t/RC}}{RC}u(t)$$

$$h(t) = e^{-at}u(t) \rightarrow 3\,dB\ \ BW = a\frac{rad}{\sec}$$

$$RC\ \ LPF \rightarrow 3dB\ \ BW = \frac{1}{RC}\frac{rad}{\sec} or \frac{1}{2\pi RC} Hz$$

Rise time (t_r):

"It is the time taken by step response of first-order C.L.S to reach from 10% to 90% of its final value".

$$t_r = 2.2RC$$

$$t_r = \frac{2.2}{3dB\,Bw\left(in\ \frac{rad}{\sec}\right)}$$

All are for 1st order systems

$$t_r = \frac{2.2}{2\pi\left(3dB\ \ Bw\ \ in\ \ Hz\right)}$$

$$t_r = \frac{0.35}{\left(3dB\ \ Bw\ \ in\ \ Hz\right)}$$

1st order system:

To analyse1st order systems,only a single parameter is needed: Time constant

2nd order system:

To analyse 2nd order systems, two parameters are needed.

parameters < ζ damping ratio; ω_n: un - damp natural frequency

1. $\zeta \rightarrow$ Damping ratio
2. $\omega_n \rightarrow$ Undamped natural frequency
3. $\zeta\omega_n \rightarrow$ Damping factor
4. $\omega_d = \omega_n\sqrt{1-\zeta^2} \rightarrow$ Damped frequency of oscillations

2nd order O.L.S.

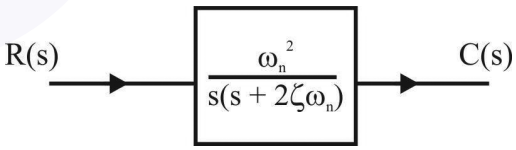


Fig. 5.24

$$T(s) = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s\left(s + 2\zeta\omega_n\right)}$$

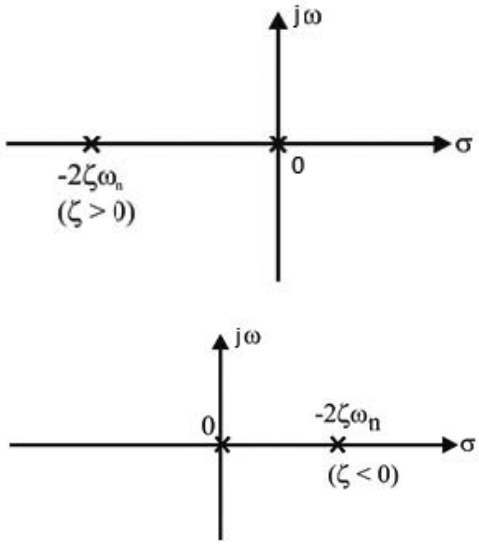


Fig. 5.25

2nd order CLS:

Case A:

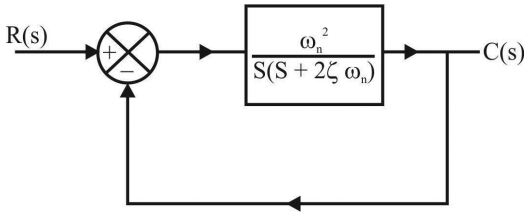


Fig. 5.26

$$T(s)=\frac{C(s)}{R(s)}=\frac{\omega_n^2}{S^2+2\zeta\omega_n S+\omega_n^2}$$

$$T(s)=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

1. $dc\ gain \rightarrow T(s=0)=1$
2. $T(s)=\frac{constant}{s^2+2\zeta\omega_n s+\omega_n^2} \rightarrow$ Standard T.F.

Case B:

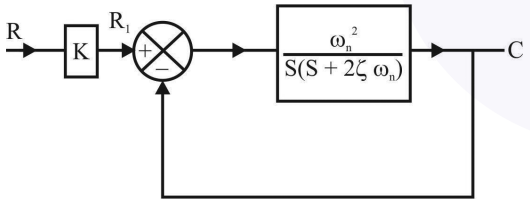


Fig. 5.27

$$\frac{C}{R}=T=\frac{k\omega_n^2}{S^2+2\zeta\omega_n S+\omega_n^2}$$

$$T=\frac{k\omega_n^2}{S^2+2\zeta\omega_n S+\omega_n^2}$$

1. D.C gain = k
2. $T(s)=\frac{constant}{S^2+2\zeta\omega_n S+\omega_n^2} \rightarrow$ standard T.F.

Case C:

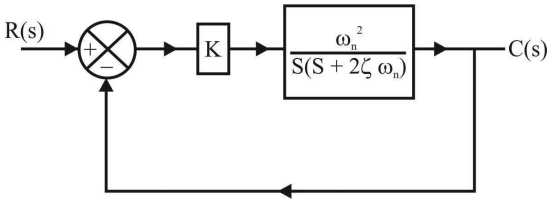


Fig. 5.28

$$\frac{C(s)}{R(s)}=\frac{k\omega_n^2}{S^2+2\zeta\omega_n S+k\omega_n^2}$$

$$T(s)=\frac{k\omega_n^2}{S^2+2\zeta\omega_n S+k\omega_n^2}$$

1. D.C gain = 1
2. $T(s)=\frac{constant}{S^2+2\zeta'\omega_n' S+(\omega_n')^2} \rightarrow$ standard from

$$\omega_n'=\omega_n\sqrt{k}$$

$$\zeta'=\frac{\zeta}{\sqrt{k}}$$

Case D:

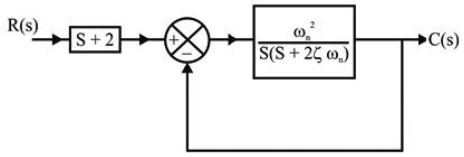


Fig. 5.29

$$\frac{C(s)}{R(s)}=\frac{(S+2)\omega_n^2}{S^2+2\zeta\omega_n S+\omega_n^2} \rightarrow \text{non standard T.F.}$$

11. The closed-loop transfer function of 2nd order system is $T(s)=\frac{C(s)}{R(s)}=\frac{20}{s^2+4s+5}$. Represent its block diagram.

Solution:

Method 1:

1. Standard T.F.
2. $T(s)=\frac{4\times5}{s^2+4s+5}=4\left\{\frac{5}{s^2+4s+5}\right\}$

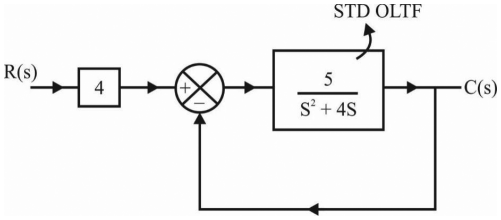


Fig. 5.30

Method 2:

$$\frac{20}{s^2+4s+5}=\frac{G(s)}{1+G(s)} \rightarrow unity-ve\ feedback$$

$$20G(s)+20=G(s)s^2+4sG(s)+5G(s)$$

$$G(s)\left[s^2+4s-15\right]=20$$

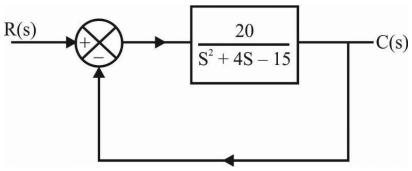


Fig. 5.31

Note:

1. For a standard 2nd order CLTF, block diagram arrangement is not unique
2. For unity –ve feedback system:

$$OLTF = \frac{CLTF}{1 - CLTF}$$

$$CLTF = \frac{OLTF}{1 + OLTF}$$

Important Definitions

1. **System damping ratio (ζ):**
   a) It is unitless (dimensionless) quantity.
   b) It describes the decay of oscillations in the output response of a system.
   c) $(-\infty < \zeta < \infty)$

2. **Undampednatural frequency $(\omega_n) \rightarrow \frac{rad}{\sec}$:**
   a) It is the frequency of oscillations of output response in the absence of damping force.
   b) $\omega_n > 0$

3. **Damped frequency of oscillations $\omega_d = \omega_n\sqrt{1-\zeta^2}\, rad / \sec$:**
   a) It is the frequency of oscillations of output response in the presence of damping force.
   b) $\omega_d > 0$

Response of 2nd order CLS with Unity(-)ve Feedback:

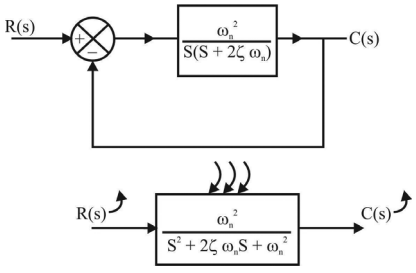
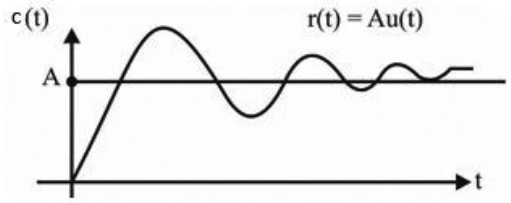


Fig. 5.32

Case 1: $0 < \zeta < 1 \rightarrow$ "*under – damp s / s*"

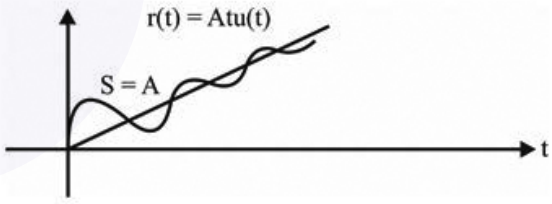
1. Output response is underdamped.
2. Output response follows the constant parameter in the input.
3. Output response nature is independent of input applied.
4. Underdamped system

a)



$c(t_s) = 0.98A$ *For* $\pm 2\%$ *error band*

b)



$$\left\{\frac{dc(t)}{dt}\right\}_{t=t_s} = 0.98A \text{ For } \pm 2\% \text{ error band}$$

c)

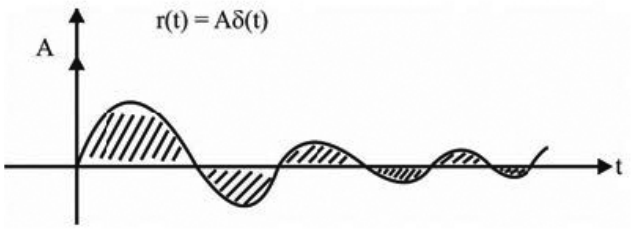


Fig. 5.33

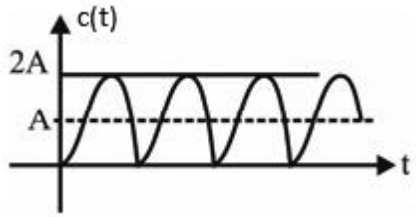
$t_s \rightarrow 2\%$ *error band*

$$\int_0 C(t)\,dt = 0.98A$$

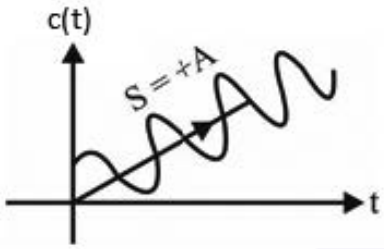
Case 2: $\zeta = 0 \rightarrow$ *un – damp s / s*

1. Output response is undamped
2. Oscillations in output response never achieve constant parameter in the input

3. Output response nature is independent of input
4. Undamped system
   a) r(t) = Au(t)



   b) r(t) = At u(t)



   c) $r(t) = A\delta(t)$

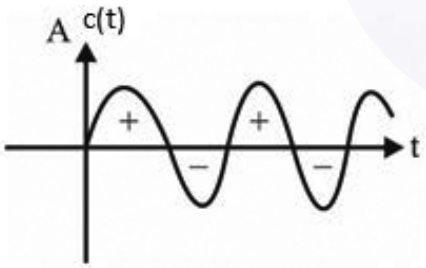
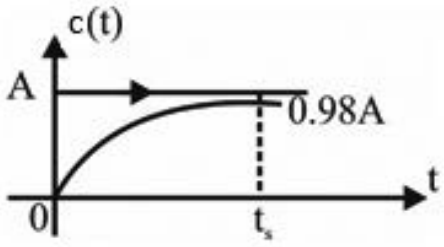


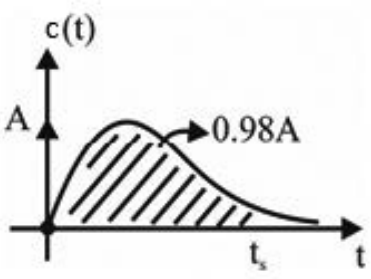
Fig. 5.34

Case 3: $\zeta \rightarrow$ *Critically damped system*

1. Output response does not oscillate and reaches a constant parameter in input in the shortest possible time.
2. Output response nature is independent of input
3. Critically dampedsystem
   a)



   b)



   c)

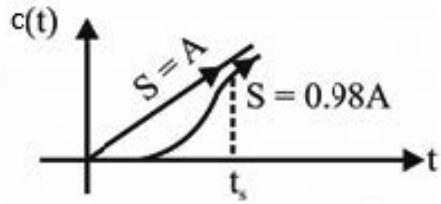
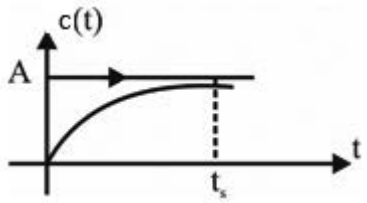


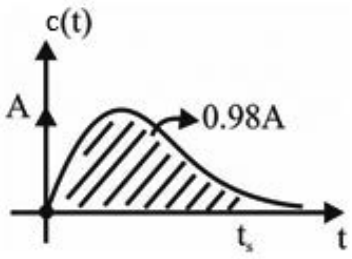
Fig. 5.35

Case 4: $\zeta > 1 \rightarrow 1 < \zeta < \infty \rightarrow$ over domped system

1. Output response does not oscillate and approaches a constant parameter of input not in the shortest possible time
2. Overdamped system
3. $A\delta(t)$

   a) Au(t)





   b) Atu(t)

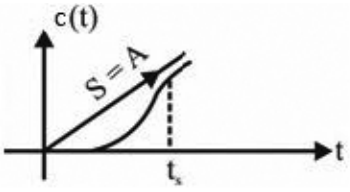


Fig. 5.36

Case 5: $-1 < \zeta < 0$

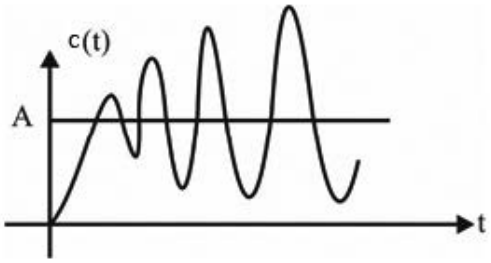


Fig. 5.36

Case 6: $\zeta = -1$

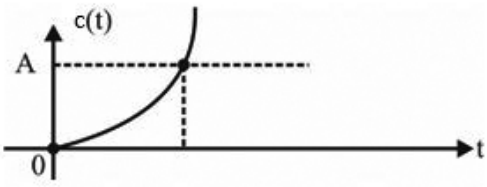


Fig. 5.37

Case 7: $\zeta < -1 : -\infty < \zeta < -1$

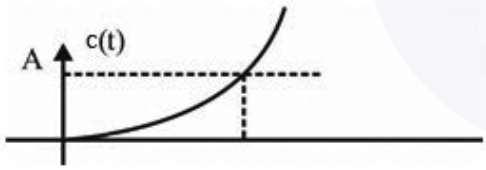


Fig. 5.38

2nd order underdamped system:

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{G(s)}{1 + G(s)H(s)} = \frac{N(s)}{D(s)}$$

1. $0 < \zeta < 1$
2. C.E. = D(s) = 1 + G(s)H(s) = 0

   $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$
3. Poles

   $$s = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2 \times 1}$$

   $s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$ $o < \zeta < 1,\ \zeta^2 - 1 < 0$

   $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \rightarrow$ *complex poles*

   $\text{Re}\{s\} = -ve \rightarrow o < \zeta < 1 \rightarrow$ *complex poles always lies on LHP*
4. Pole-zero plot:

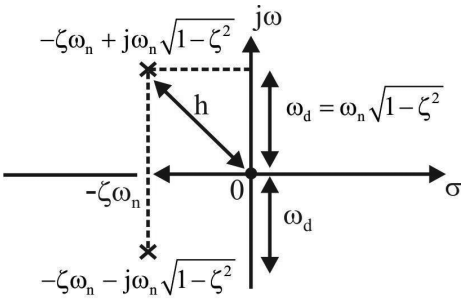


Fig. 5.39

Pole location: Variable

$h^2 = \zeta\omega_n^2 + \omega_d^2$

$h = \omega_n \rightarrow$ constant

Time constant:

$$T = \frac{1}{\zeta\omega_n}$$

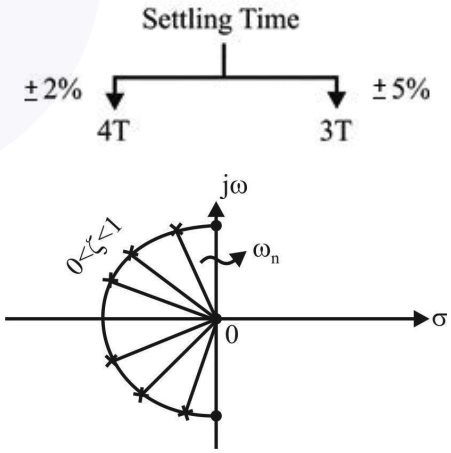


Fig. 5.40

Note: As $0 < \zeta < 1 \rightarrow$ Complex poles lie on a semicircle of radius ω_n

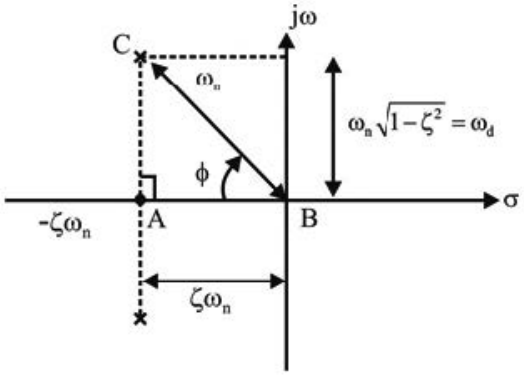


Fig. 5.41

ΔABC

$\cos\phi = \zeta$

$0 < \zeta < 1$

Summary

1. Poles of the underdamped system lie on L.H. of s-plane with –ve real part
2. Underdamped systems are always stable.
3. $0 < \zeta < 1 \rightarrow$ poles move on semicircle of radius ω_n.

2nd Order Undamped System:

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

1. $\zeta = 0$
2. $C.E. = 1 + G(s)H(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$
   $= s^2 + \omega_n^2 = 0$
3. Pole locations, $s = \pm j\omega_n$
4. Pole-zero diagram
   Pole location is constant

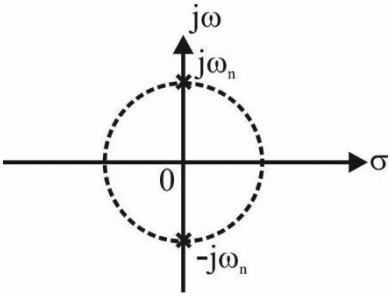


Fig. 5.42

Time constant $T = \frac{1}{0} = \infty$, *settling time* $\rightarrow \infty$

Summary

1. Poles of 2nd order undamped system lie on $j\omega$-axis.
2. 2nd order undamped systems are marginally stable.
3. Poles of 2nd order undamped system also lie on a circle of radius ω_n.

2nd order critically damped system:

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^s}$$

1. $\zeta = 1$
2. $1 + G(s)H(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$
   $= s^2 + 2\omega_n s + \omega_n^2 = (s + \omega_n)^2$
3. Poles:
   $s = -\omega_n, -\omega_n \rightarrow$ Real, Negative, Repeated
4. Pole-zero location
   Pole location is fixed

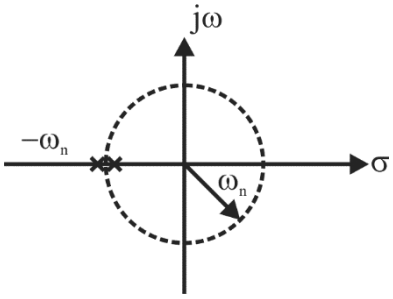


Fig. 5.43

Time constant:

$$T = \frac{1}{\omega_n}$$

Settling time = 6T
(2% error band)

Summary

1. Poles of 2nd order critically dampedsystem are –ve, real and repeated.
2. Critically dampeds ystems are always stable.
3. Poles of 2nd order critically dampedsystem lie on the circle of radius ω_n

2nd order overdamped system:

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

1. $\zeta > 1$ or $1 < \zeta < \infty$
2. $1 + G(s)H(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$

   $$s = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2 \times 1} \qquad \zeta > 1$$

   $$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \qquad \zeta^2 - 1 > 0$$

3. Poles: $s = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$

   $s = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$ (-)ve real but distinct.

$$-\zeta\omega_n + \underbrace{\sqrt{\zeta^2\omega_n^2 - \omega_n^2}}_{<\zeta\omega_n} \quad \sqrt{\zeta^2\omega_n^2 - \omega_n^2} < \zeta\omega_n$$

4. Pole-zero location

   Pole locations are variable

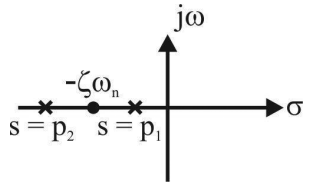


Fig. 5.44

$$T = \frac{1}{p_1}$$

$$t_s = 4T \text{ for 2\% error band}$$

Summary

1. Pole locations of 2nd order overdamped system arereal, -ve and distinct.
2. Poles' locations do not lie on a circle of radius ω_n.
3. 2nd order overdamped systems are always stable.

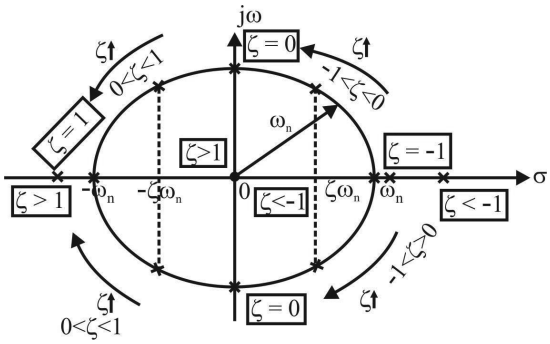


Fig. 5.46

Note:

If $\zeta < 0$

1. The systems are unstable.
2. The outputs are unbounded.

Summary

Nature of System	ζ	Nature of Response	Time Constant τ	2% t_s	Stability
Under damped	$0 < \zeta < 1$	Damped oscillatory	$\frac{1}{\zeta\omega_n}$	4τ	Stable
Undamped	$\zeta = 0$	Undamped oscillatory	∞	∞	Marginally
Critically Damped	$\zeta = 1$	No oscillations, fast settling	$\frac{1}{\omega_n}$	6τ	Stable
Overdamped	$1 < \zeta < \infty$	No oscillations, slow settling	$\frac{1}{\lvert\text{dominant Pole}\rvert}$	4τ	Stable

Table 5.1

Step responses:

Step response of underdamped system:

$(0 < \zeta < 1)$

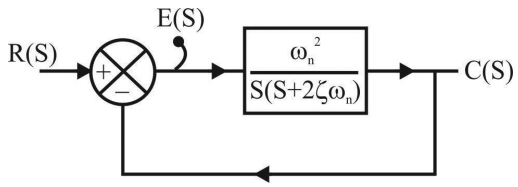


Fig. 5.45

$$E(S) = R(S) - C(S) \rightarrow e(t) = r(t) - c(t)$$

$$T(S) = \frac{\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2}$$

$$r(t) = Au(t) \leftrightarrow R(s) = \frac{A}{S}$$

Step response:

$$C(S) = R(S)T(S) = \frac{A\omega_n^2}{S\left(S^2 + 2\zeta\omega_n S + \omega_n^2\right)}$$

$\rightarrow$ *in S domain*

$$C(S) = \frac{A\omega_n^2}{S\left(S-\left(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}\right)\right)\left(S-\left(-\zeta\omega_n - j\omega_n\sqrt{1-\zeta^2}\right)\right)}$$

$$C(S) = \frac{a_0}{S} + \frac{a_1}{S-(-a+jb)} + \frac{a_2}{\left(S-(-a+jb)\right)} \quad \updownarrow \text{ I.L.T.}$$

$$c(t) = A\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \phi\right)\right\}u(t)$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\cos\phi = \zeta \rightarrow \phi = \cos^{-1}\zeta$$

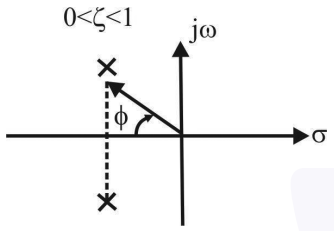


Fig. 5.47

$$e(t) = r(t) - c(t)$$

$$e(t) = Au(t) - c(t) = A\left\{\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \phi\right)\right\}u(t)$$

1. Time instant of A
2. Time instant of maxima
3. Time instants of minima

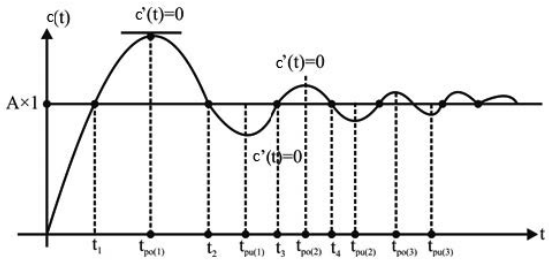


Fig. 5.48

Calculation of 't_k' k = 1, 2, 3

c(t) = A

$$A\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \phi\right)\right\} = A$$

$$1 - \frac{e^{-\zeta\omega_n t}\sin\left(\omega_d t + \phi\right)}{\sqrt{1-\zeta^2}} = 1$$

$$\frac{e^{-\zeta\omega_n t}\sin\left(\omega_d t + \phi\right)}{\sqrt{1-\zeta^2}} = 0$$

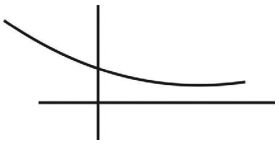


Fig. 5.49

$$\sin\left(\omega_d t + \phi\right) = 0$$

$$\sin\left(\omega_d t + \phi\right) = \sin n\pi$$

$$\omega_d t + \phi = n\pi$$

$$t = \frac{n\pi - \phi}{\omega_d}$$

$n \rightarrow$ positive integer such that t is +ve

n = 1, 2, 3 .....

$$t_1 = \frac{\pi - \phi}{\omega_d}$$

$$t_2 = \frac{2\pi - \phi}{\omega_d}$$

$$t_3 = \frac{3\pi - \phi}{\omega_d}$$

$$t_2 - t_1 = \frac{\pi}{\omega_d}$$

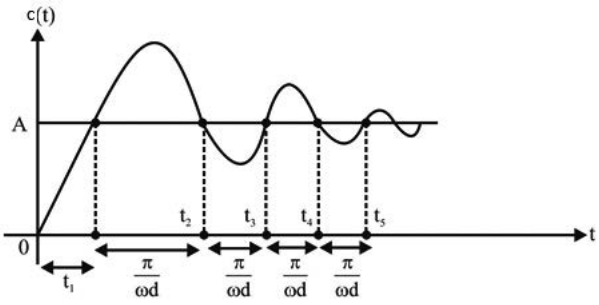


Fig. 5.50

$$t_1 = \frac{\pi - \phi}{\omega_d}$$

Rise time:

"It is the time taken by the output (step) response of 2nd order underdampedsystem to reach from 0% to 100% of its final value or steady-state value for the very first time".

$$t_r = \frac{\pi - \phi}{\omega_d} = \frac{\pi - \phi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi - \cos^{-1}\zeta}{\omega_n\sqrt{1-\zeta^2}}$$

Note: $\pi = 3.14$ ϕ = radians

Peak times:

1. Peak overshoot time: $\frac{dc(t)}{dt} = 0$

2. Peak undershoot time:

$$c(t) = A\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \phi\right)\right\}$$

$$\frac{dc(t)}{dt} = 0$$

$$-A\left\{\begin{array}{l}\dfrac{e^{-\zeta\omega_n t}\left(-\zeta\omega_n\right)\sin\left(\omega_d t + \phi\right)}{\sqrt{1-\zeta^2}} \\ + \dfrac{e^{-\zeta\omega_n t}\omega_d\cos\left(\omega_d t + \phi\right)}{\sqrt{1-\zeta^2}}\end{array}\right\} = 0$$

$$+\zeta\omega_n\sin\left(\omega_d t + \phi\right) = \omega d\ \cos\left(\omega_d t + \phi\right)$$

$$\tan\left(\omega_d t + \phi\right) = \frac{\omega_d}{\zeta\omega_n} \times \frac{1}{\zeta\omega_n} \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\tan\left(\omega_d t + \phi\right) = \frac{\omega_n\sqrt{1-\zeta^2}}{\omega_n\zeta}\ ; \qquad \cos\phi = \zeta$$

$$\tan\left(\omega_d t + \phi\right) = \frac{\sqrt{1-\zeta^2}}{\zeta} = \tan\phi\,;$$

$$\tan\phi = \sqrt{\frac{1-\zeta^2}{\zeta}}$$

$$\tan\left(\omega_d t + \phi\right) = \tan\left(n\pi + \phi\right)\ \ \sin\alpha = \sin\beta$$

$$\omega_d t + \phi = n\pi + \phi \quad \alpha = \beta + 2n\pi$$

$$t = \frac{n\pi}{\omega_d}\ n \rightarrow Integer\ \ such\ that\ \ t\ \ is\ \ +ve$$

$$n = 1, 3, 5\ldots \rightarrow t_{po(k)}$$

$$= \frac{(2k-1)\pi}{\omega_d} \quad k = 1, 2, 3\ldots$$

$$n = 2, 4, 6\ldots.. \rightarrow t_{pu(k)} = \frac{2\pi k}{\omega_d} \quad k = 1, 2, 3\ldots.$$

Peak overshoot time:

1. It occurs in each cycle: Hence it is periodic/cyclic in nature.
2. At peak overshoot time, $\frac{dc(t)}{dt} = 0$
3. At first peak overshoot time, the value of output is maximum
4. $c\left(t - t_{po(1)}\right) = \left\{c(t)\right\}_{\max}$
5. Time gap between two successive P.O. is $\frac{2\pi}{\omega_d}$

Peak undershoot time:

1. This occurs in each cycle
2. At this time, $\frac{dc(t)}{dt} = 0$
3. First peak undershoot time →Time period of damped oscillation

$$T_D = \frac{2\pi}{\omega_d} = t_{pu(1)}$$

4. At first peak undershoot time, the output is minimum after rising time.
5. Time gap between successive undershoots $= \frac{2\pi}{\omega_d}$

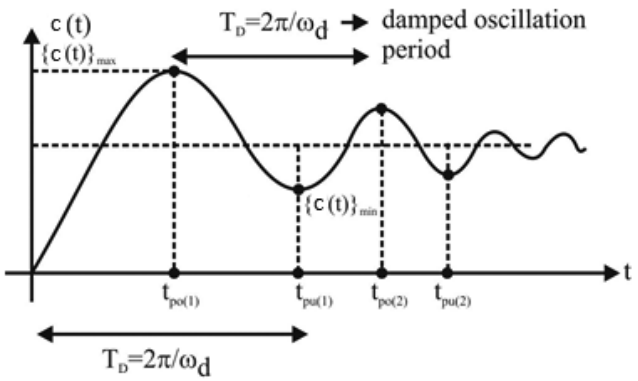


Fig. 5.51

Concept of overshoot and undershoot amplitude:

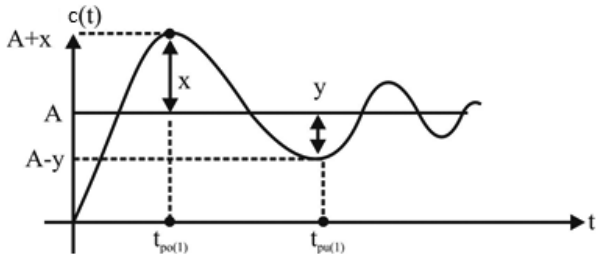


Fig. 5.52

$$c(t) = A\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \phi\right)\right\}$$

$$\left\{c(t)_{ss} = \lim_{t\to\infty} c(t) = A\right\}$$

1. Overshoot = $c(t) - \left\{c(t)\right\}_{ss}$
2. Peak overshoot = $c\left(t_{po(k)}\right) - \left\{c(t)\right\}_{ss}$
3. Maximum peak overshoot

   $= c\left(t_{po(1)}\right) - \left\{c(t)\right\}_{ss}$

   $= \left\{c(t)\right\}_{max} - \left\{c(t)\right\}_{ss}$
4. Percentage peak overshoot

   $$= \frac{\left\{c\left(t_{po(k)}\right)\right\} - \left\{c(t)\right\}_{ss}}{\left\{c(t)\right\}_{ss}} \times 100\%$$
5. Percentage maximum peak overshoot

   $$= \frac{c(t)_{max} - \left\{c(t)\right\}_{ss}}{\left\{c(t)\right\}_{ss}} \times 100\%$$

1. Undershoot = $\left\{c(t)\right\}_{ss} - c(t)$
2. Peak undershoot = $\left\{c(t)\right\}_{ss} - c\left(t_{pu(k)}\right)$
3. Maximum peak undershoot

   $= \left\{c(t)\right\}_{ss} - c\left(t_{pu(1)}\right)$
4. % Peak undershoot

   $$= \left\{\frac{c(t)_{ss} - c\left(t_{pu(k)}\right)}{c(t)_{ss}}\right\} \times 100\%$$
5. Maximum % peak undershoot

   $$= \frac{\left\{c(t)\right\}_{ss} - \left(c(t)\right)_{min}}{\left\{c(t)\right\}_{ss}} \times 100\%$$

Calculation of maximum peak values:

Maximum peak overshoot

$$= c\left(t_{po(k)}\right) - \left\{c(t)\right\}_{ss}$$

$$t_{po(k)} = \frac{(2k-1)\pi}{\omega_d}$$

$$c(t) = A\frac{\left\{1 - e^{-\zeta\omega_n t}\sin\left(\omega_d t + \phi\right)\right\}}{\sqrt{1-\zeta^2}}$$

$$c(t) = A\left\{1 - \frac{e^{\frac{-\zeta\omega_n(2k-1)\pi}{\omega_n\sqrt{1-\zeta^2}}}\sin\left((2k-1)\pi + \phi\right)}{\sqrt{1-\zeta^2}}\right\}$$

$$c(t) = A\left\{1 + \frac{e^{-(2k-1)\pi\zeta/\sqrt{1-\zeta^2}}\sin\phi}{\sqrt{1-\zeta^2}}\right\} \qquad \cos\phi = \zeta \quad \sin\phi = \sqrt{1-\zeta^2}$$

$$c(t) = A\left\{1 + e^{-(2k-1)\pi\zeta/\sqrt{1-\zeta^2}}\right\}$$

$k = 1$ $\qquad \left\{c(t)\right\}_{max} = A\left\{1 + e^{-\pi\zeta/\sqrt{1-\zeta^2}}\right\}$

$k = 2$ $\qquad \left\{c(t)\right\} = A\left\{1 + e^{-3\pi\zeta/\sqrt{1-\zeta^2}}\right\}$

Maximum peak overshoot:

$$M_{PO(1)} = \left\{c(t)\right\}_{max} - \left\{c(t)\right\}_{ss} = Ae^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

$$M_{PO(1)} = Ae^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

% Maximum peak overshoot

$$= \%M_{PO(1)} = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$$

Important:

$\%M_p = m\%$ $\qquad$ *calculate* $\zeta = ?$

S.1: $p = \frac{m}{100}$

s. 2: $\zeta = \sqrt{\dfrac{(\ln p)^2}{\pi^2 + (\ln p)^2}}$

12. $\%M_p = 16\%$ *calculate* ζ

Solution:

$$\%M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = 16\%$$

$$p = \frac{16}{100} = 0.16 \rightarrow \zeta = \sqrt{\frac{(\ln p)^2}{\pi^2 + (\ln p)^2}} = 0.5$$

Maximum peak undershoot

Peak undershoot = $c(t)\big|_{ss} - c(t)\big|_{t=t_{pu(k)}}$

$$C\left(t = t_{pu(k)}\right) = A\left(1 - e^{-2k\pi\zeta/\sqrt{1-\zeta^2}}\right) \quad t_{pu(k)} = \frac{2k\pi}{\omega_d}$$

Peak undershoot = $Ae^{-2\pi k\zeta/\sqrt{1-\zeta^2}}$

Maximum peak undershoot = $Ae^{-2\pi\zeta/\sqrt{1-\zeta^2}}$ (for k =1)

% Maximum peak undershoot

$= e^{-2\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$

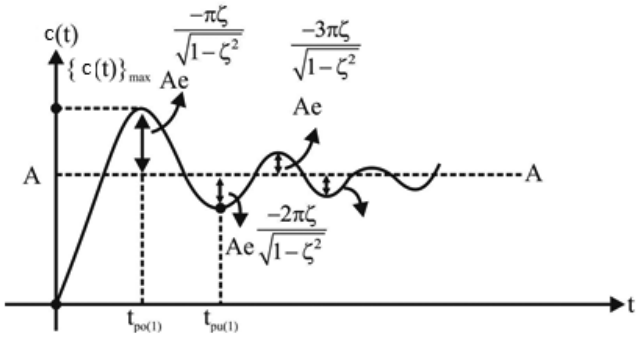


Fig. 5.53

Points To Remember

1. Maximum values of c(t)

$$\{c(t)\}_{maxs} = A\left\{1 + e^{\frac{-(2k-1)\pi\zeta}{\sqrt{1-\zeta^2}}}\right\}$$

2. Height of overshoot = $Ae^{\frac{-(2k-1)\pi\zeta}{\sqrt{1-\zeta^2}}}$

3. Minimum values of c(t)

$$\{c(t)\}_{min\,s} = A\left(1 - e^{\frac{-2k\pi\zeta}{\sqrt{1-\zeta^2}}}\right)$$

4. Height of undershoots: $Ae^{-(2k\pi)\zeta/\sqrt{1-\zeta^2}}$

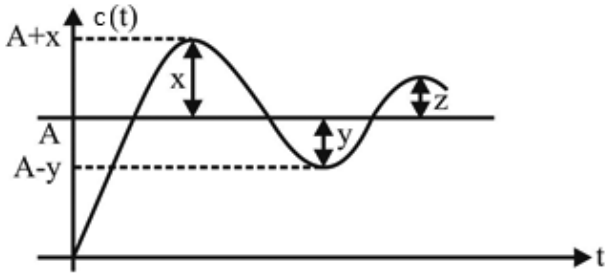


Fig. 5.54

$$\frac{Z}{A} \times 100\% = e^{-3\pi\zeta/\sqrt{1-\zeta^2}} \times 100$$

$$\frac{(A+x)-A}{A} \times 100\% = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = \frac{\{C(t)\}_{max} - \{C(t)\}_{ss}}{\{C(t)\}_{ss}}$$

$$\frac{A-(A-y)}{A} \times 100\% = e^{-2\pi\zeta/\sqrt{1-\zeta^2}} \times 100\% = \frac{\{C(t)\}_{ss} - \{C(t)\}_{min}}{\{C(t)\}_{ss}}$$

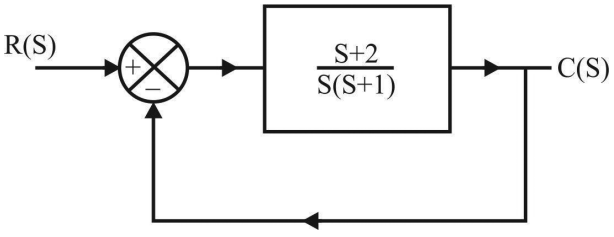


Fig. 5.55

13. Calculate the maximum peak percentage overshoot for unit step response of the above system.

Solution: $\dfrac{C(S)}{R(S)} = \dfrac{S+2}{S^2+2S+2}$

$$r(t) = u(t) \leftrightarrow R(s) = \frac{1}{S}$$

$$C(S) = \frac{(S+2)}{S(S^2+2S+2)}$$

$$C(S) = \frac{S+2}{S(S^2+2S+2)} = \frac{a_0}{S} + \frac{a_1 S + a_2}{S^2+2S+2}$$

$$C(S) = \frac{1}{S} - \frac{S+1}{S^2+2S+2}$$

$$C(S) = \frac{1}{S} - \frac{(S+1)}{(S+1)^2+1^2}$$

$$c(t) = \left\{1 - e^{-t}\cos t\right\}$$

$$\%M_P = \frac{\{c(t)\}_{max} - \{c(t)\}_{ss}}{\{c(t)\}_{ss}}$$

$$c(t)\Big|_{ss} = \lim_{t\to\infty} c(t) = 1$$

$$c(t)\Big|_{max} \rightarrow \frac{dc(t)}{dt} = 0$$

$$-\left\{-e^{-t}\sin t + \cos t\left(-e^{-t}\right)\right\} = 0$$

$$-e^{-t}\left[\sin t + \cos t\right] = 0$$

$$\tan(t) = -1 \rightarrow t = (4n+1)\frac{\pi}{4} \text{ or } (4n-1)\frac{\pi}{4}$$

$$t = \frac{-\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4} \ldots\ldots\ldots$$

$$t_{po(1)} = \frac{3\pi}{4}$$

$$\{c(t)\}_{max} = c\left(t = \frac{3\pi}{4}\right)$$

$$= \left\{1 - e^{\frac{3\pi}{4}}\cos\left(\frac{3\pi}{4}\right)\right\} = 1.067$$

$$\%M_P = \frac{1.067 - 1}{1} \times 100\% = 6.7\%$$

Common mistake:

1. Students directly try to use the formula for % overshoot or undershoot without observing the transfer function.
2. Do not directly use the formula for any calculation. Use a standard approach for any type of typical system.

Summary

Case A: $\frac{C(S)}{R(S)} = \frac{\omega_n^2}{S^2 + 2\zeta\omega_n s + \omega_n^2} \quad (0 < \zeta < 1)$

1. Standard T.F.
2. r(t) = Au(t)
3. $c(t) = A\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right\}$
4. $e(t) = \left\{\frac{Ae^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right\}$
5. $t_{po(k)} = \frac{(2k-1)\pi}{\omega_d} \qquad k = 1, 2.....$
6. $t_{pu(k)} = \frac{2k\pi}{\omega_d} \qquad k = 1, 2.....$
7. $t_{pu(1)} = \frac{2\pi}{\omega_d} = T_D$

   T_D : *Time period of damped oscillation*

   ω_d : *frequency of damped oscillation*
8. $t_s = 4T = \frac{4}{\zeta\omega_n} \rightarrow \pm 2\,\%\,error\ band$
9. No. of cycles in oscillation, $n = \frac{t_s}{T_D}$
10. Output maxima's $\rightarrow c(t) = A\left\{1 + e^{\frac{-(2k-1)\pi\zeta}{\sqrt{1-\zeta^2}}}\right\}$
11. Output minima's $\rightarrow c(t) = A\left\{1 - e^{\frac{-2k\pi\zeta}{\sqrt{1-\zeta^2}}}\right\}$
12. Peak overshoot length = $Ae^{\frac{-(2k-1)\pi\zeta}{\sqrt{1-\zeta^2}}}$
13. Peak undershoot length = $Ae^{\frac{-2k\pi\zeta}{\sqrt{1-\zeta^2}}}$
14. Maximum peak overshoot length = $Ae^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$
15. Maximum peak undershoot length

    = $Ae^{\frac{-2\pi\zeta}{\sqrt{1-\zeta^2}}}$

16. Maximum peak % overshoot,

$$\%M_P = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$$

17. $\zeta = \sqrt{\dfrac{(\ln p)^2}{\pi^2 + (\ln p)^2}}$; $p = \dfrac{m}{100}$

18. Rise time: $t_r = \dfrac{\pi - \phi}{\omega_d}$

Case B:

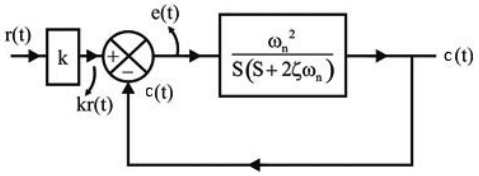


Fig. 5.16

$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2}$$

1. Standard T.F.
2. r(t) = Au(t)
3. $c(t) = Ak\left\{1 - \dfrac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right\}$
4. $e(t) = \dfrac{Ake^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)$
5. $t_{pu(k)} = \dfrac{(2k\pi)}{\omega_d}$
6. $t_{po(k)} = \dfrac{(2k-1)\pi}{\omega_d}$
7. $t_{pu(1)} = \dfrac{2\pi}{\omega_d} = T_D$
8. $t_s = \dfrac{4}{\zeta\omega_n}$
9. $n = \dfrac{t_s}{T_D}$
10. *Output* maxima's $\rightarrow c(t) = Ak\left(1 + e^{\frac{-(2k-1)\pi\zeta}{\sqrt{1-\zeta^2}}}\right)$
11. *Output* minima's $\rightarrow c(t) = Ak\left(1 - e^{\frac{-2k\pi\zeta}{\sqrt{1-\zeta^2}}}\right)$
12. *Peak over shoots length* $= Ake^{\frac{-(2k-1)\pi\zeta}{\sqrt{1-\zeta^2}}}$
13. *Peak under shoots length* $= Ake^{\frac{-2k\pi\zeta}{\sqrt{1-\zeta^2}}}$
14. *Maximum peak over shoot length*

$= Ake^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$

15. *Maximum peak under shoot length*

$= Ake^{\frac{-2\pi\zeta}{\sqrt{1-\zeta^2}}}$

16. $\%M_P = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$

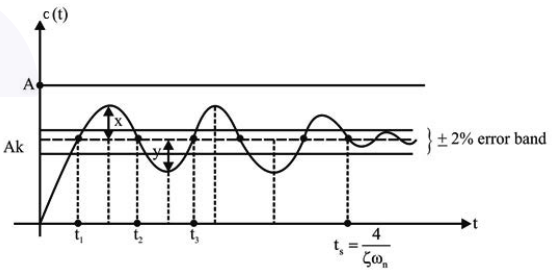


Fig. 5.56

Test Your Knowledge

1.

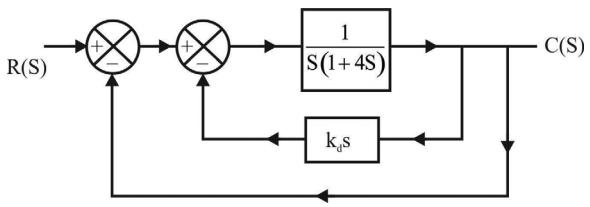


Fig. 5.57

$k_d = ?$ *such that* $\zeta = 0.5$

Solution: $k_d = 1$

2.

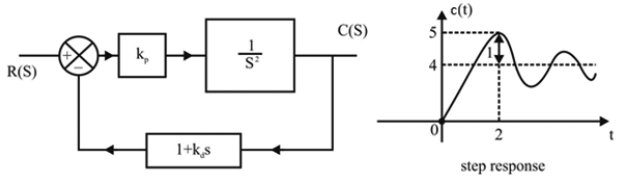


Fig. 5.58

Calculate: k_p *and* k_d = ?

Solution: $k_p = 2.944$

$k_d = 0.468$

Step response of undamped system:

$\zeta = 0$

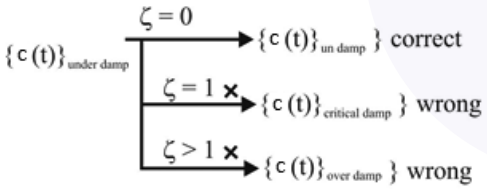
$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2}$$

$$r(t) = Au(t) \leftrightarrow R(S) = A/S$$

$$C(S) = \frac{Ak\omega_n^2}{S\left(S^2 + \omega_n^2\right)} = \frac{a_0}{S} + \frac{a_1 S + a_2}{S^2 + \omega_n^2}$$

$$c(t) = Ak\left(1 - \cos\omega_n t\right)u(t)$$

$$\phi = \cos^{-1}\zeta = \cos^{-1}0 = \frac{\pi}{2}$$



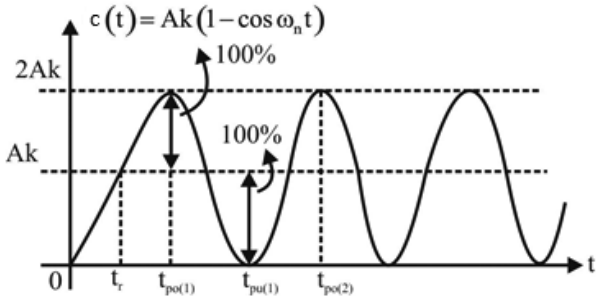


Fig. 5.59

1. $t_r = \dfrac{\pi - \phi}{\omega_d} \xrightarrow{\zeta=0} \dfrac{\pi - \frac{\pi}{2}}{\omega_n} = \dfrac{\pi}{2\omega_n}$

   $\omega_d = \omega_n\sqrt{1-\zeta^2}$

2. $t_{po(1)} = \dfrac{(2k-1)\pi}{\omega_d} \xrightarrow{\zeta=0} \dfrac{(2k-1)\pi}{\omega_n}$

3. $t_{pu(1)} = \dfrac{2k\pi}{\omega_n}$

4. $t_s = \infty$

5. $\%M_P = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = 100\%$

Points To Remember

Calculate frequency of oscillations of output:

$0 < \zeta < 1 \rightarrow \omega_d = \omega_n\sqrt{1-\zeta^2}$

$\zeta = 0 \rightarrow \omega_n$

Step response of critically damped system:

$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2} \qquad \zeta = 1$$

$$r(t) = Au(t) \leftrightarrow R(S) = \frac{A}{S}$$

$$C(S) = \frac{Ak\omega_n^2}{S\left(S+\omega_n\right)^2} = \frac{a_0}{S} + \frac{a_1}{\left(S+\omega_n\right)} + \frac{a_2}{\left(S+\omega_n\right)^2}$$

$$C(S) \xleftrightarrow{I.L.T} c(t)$$

$$c(t) = Ak\left(1 - e^{-\omega_n t}\left(1 + \omega_n t\right)\right)u(t)$$

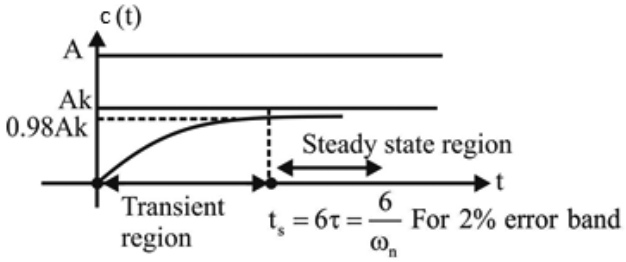


Fig. 5.60

Keywords:

1. Poles of CLS are real, (-)ve and equal.
2. Output response reaches to steady-state without oscillations in the shortest possible time.

Step response of overdamped system: $\zeta > 1$

$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2} = \frac{k\omega_n^2}{(S+p_1)(S+p_2)}$$

$$r(t) = Au(t) \leftrightarrow R(S) = A/S$$

$$C(S) = \frac{Ak\omega_n^2}{(S+p_1)(S+p_2)S} = \frac{a_0}{S} + \frac{a_1}{(S+p_1)} + \frac{a_2}{(S+p_2)}$$

$$c(t) = \left\{a_0 + a_1 e^{-p_1 t} + a_2^{-p_2 t}\right\} u(t)$$

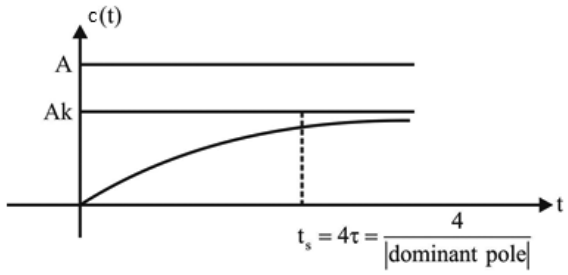


Fig. 5.64

Important Points

1.

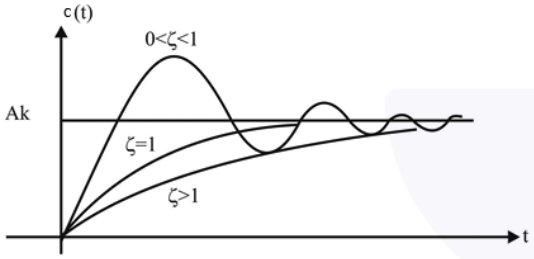


Fig. 5.61

2.

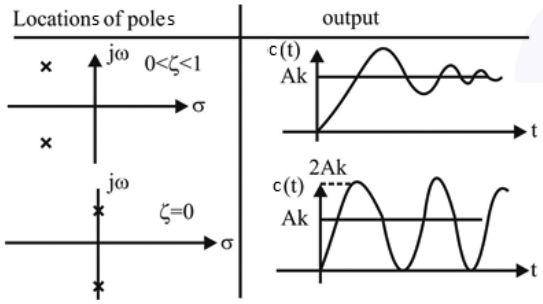


Fig. 5.62

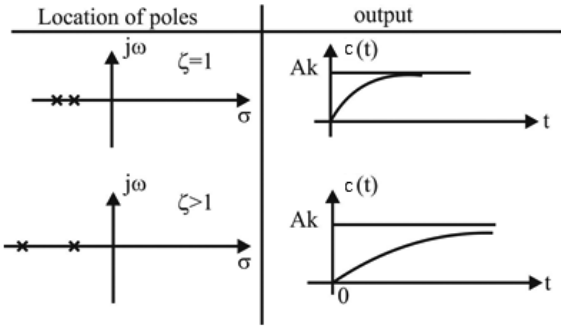


Fig. 5.63

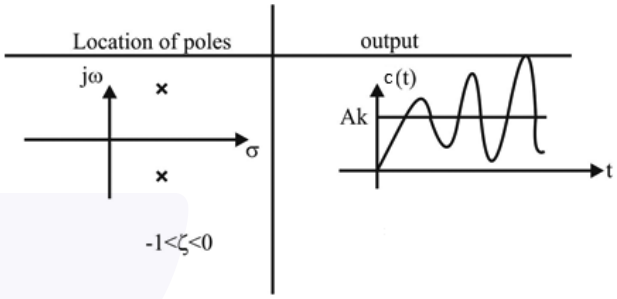


Fig. 5.65

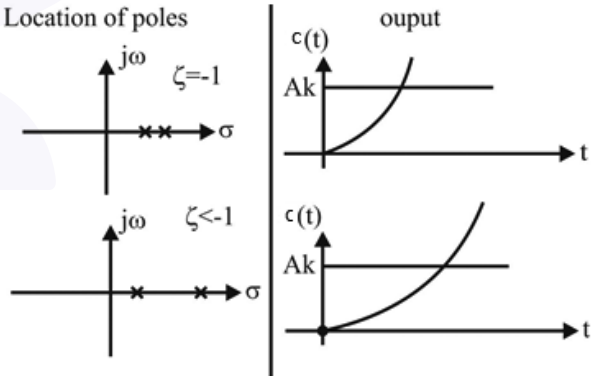


Fig. 5.66

3.

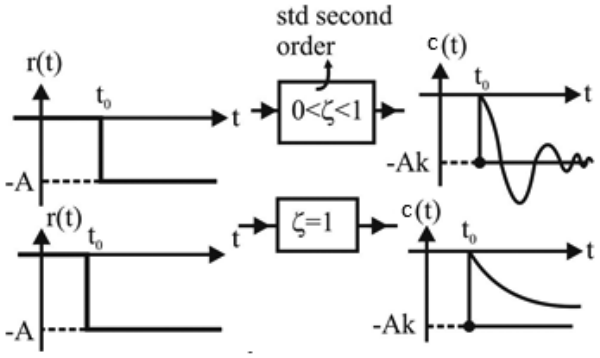


Fig. 5.67

4. Locus diagrams $(0 < \zeta < 1)$

a) Constant ζ locus

$\cos\phi = \zeta$ → ζ = constant → ϕ = constant

But ω_n is variable:

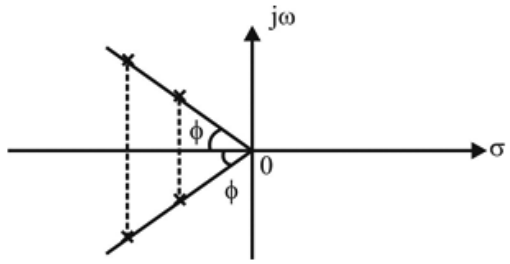


Fig. 5.68

b) Constant ω_d locus

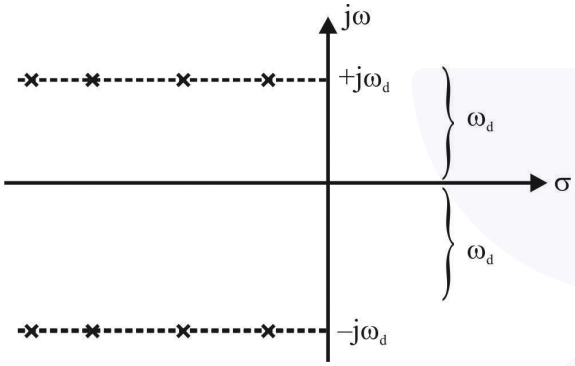


Fig. 5.69

$\omega_n \rightarrow$ variable

$\zeta \rightarrow$ variable

$\omega_d = \omega_n\sqrt{1-\zeta^2}$ = Constant = k

$$\omega_n = \frac{k}{\sqrt{1-\zeta^2}} = \frac{k}{\sin\phi}$$

c) Constant ω_n locus

$\zeta \rightarrow$ variable

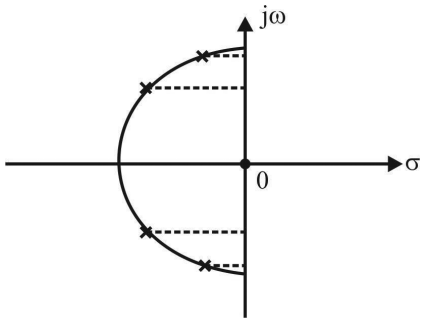


Fig. 5.70

d) Constant time constant (Settling time) locus

$\zeta\omega_n \rightarrow$ constant

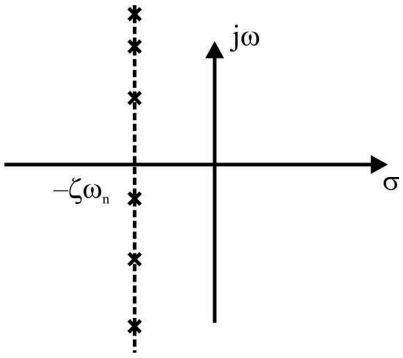


Fig. 5.71

Impulse response of 2nd order system:

1. Impulse response of underdamped system: $(0 < \zeta < 1)$

$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2}$$

$$r(t) = A\delta(t) \leftrightarrow R(S) = A$$

$$C(S) = \frac{Ak\omega_n^2}{\left(S^2 + 2\zeta\omega_n S + \omega_n^2\right)}$$

$$= \frac{Ak\omega_n^2}{\left(S - \left(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}\right)\right)\left(S - \left(-\zeta\omega_n - j\omega_n\sqrt{1-\zeta^2}\right)\right)}$$

$$c(t) = \frac{Ak\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin\left(\omega_d t\right)u(t) \rightarrow h(t)$$

$$s(t) = Ak\left\{1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \phi\right)\right\}u(t)$$

$$\frac{ds(t)}{dt} = h(t)$$

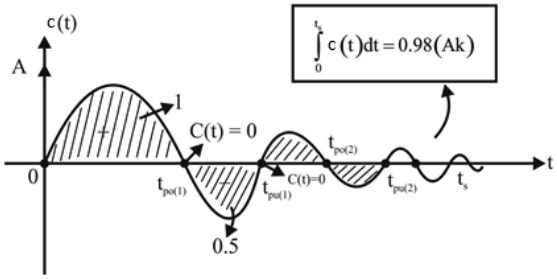


Fig. 5.72

$t_s = 4\tau \rightarrow \pm 2\%\, error\ band$

1. $t_r = \dfrac{\pi - \phi}{\omega_d}$ Procedure: $\int_{-\infty}^{\infty} c(t)\,dt = Ak$

$t_m = \dfrac{m\pi - \phi}{\omega_d}$

2. $t_{po(k)} = \dfrac{(2k-1)\pi}{\omega_d}$

$t_{pu(k)} = \dfrac{2k\pi}{\omega_d}$

Procedure: $g(t) = \int_{\infty}^{t} c(t)\,dt \rightarrow Running\ Area$

$\dfrac{dg(t)}{dt} = 0 \rightarrow t \begin{cases} t_{po} \\ t_{pu} \end{cases}$

$c(t) = 0 \nearrow$

3. $t_s = \dfrac{4}{\zeta\omega_n}$

4. $\%M_P = \dfrac{\int_0^{t_{po(1)}} c(t)\,dt - \int_0^{t_s} c(t)\,dt}{\int_0^{t_s} c(t)\,dt}$

$\times 100\% = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$

2. Impulse response of 2nd order undamped system:

$\dfrac{C(S)}{R(S)} = \dfrac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2} \qquad \zeta = 0$

$r(t) = A\delta(t) \leftrightarrow R(S) = A$

$C(S) = \dfrac{Ak\omega_n^2}{S^2 + \omega_n} \leftrightarrow c(t) = Ak\omega_n \sin(\omega_n t)u(t)$

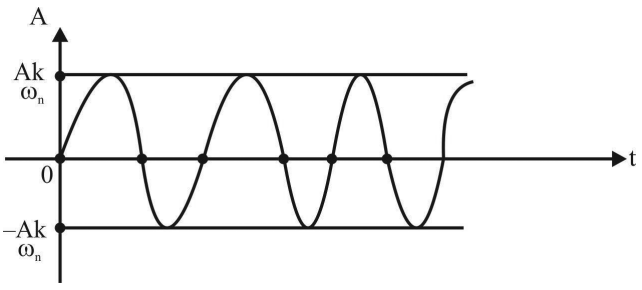


Fig. 5.73

3. Impulse response of 2nd order critical damped system:

$\dfrac{C(S)}{R(S)} = \dfrac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2} \qquad (\zeta = 1)$

$r(t) = A\delta(t)$

$C(S) = \dfrac{Ak\omega_n^2}{(S+\omega_n)^2} \xleftrightarrow{I.L.T.} c(t)$

$= Ak\omega_n^2 t\, e^{-\omega_n t} u(t)$

$\int_0^{t_s} c(t)\,dt = 0.98Ak$

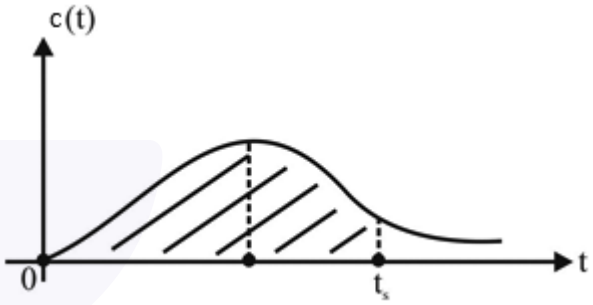


Fig. 5.74

$t_s = 6T = \dfrac{6}{\omega_n}\, 2\%\, error\ band$

4. Impulse response of 2nd order over-damped s/s:

$\dfrac{C(s)}{R(s)} = \dfrac{k\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2}$

$r(t) = A\delta(t) \leftrightarrow R(S) = A$

$C(S) = \dfrac{Ak\omega_n^2}{(S+p_1)(S+p_2)}$

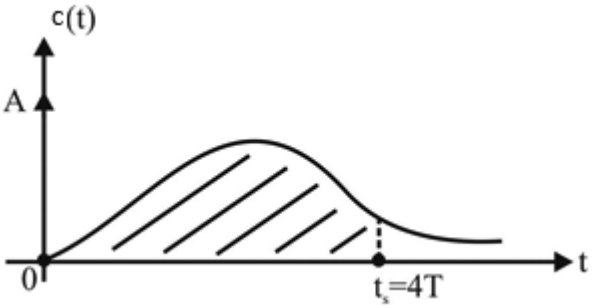


Fig. 5.75

Ramp response of 2nd order system:

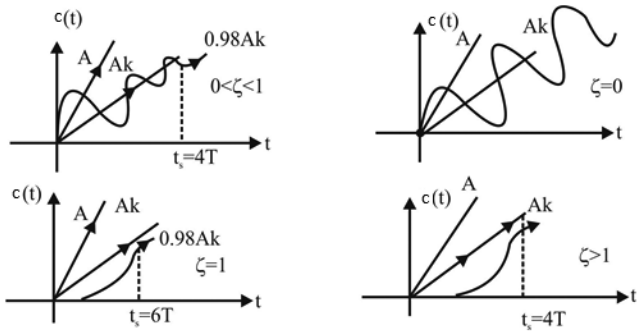


Fig. 5.76

Dominant pole concept:

$$\frac{C(S)}{R(S)} = \frac{M}{(S+S_1)(S+S_2)(S+S_3)(S+S_4)}$$

Consider $p_1, p_2, p_3, p_4 \rightarrow$ Magnitude of real part of poles.

Smallest of $(p_1, p_2, p_3, p_4) = p_1$

$$\frac{p_2}{p_1} \geq 5 \rightarrow (S+S_2) \qquad \frac{p_3}{p_1} \geq 5 \rightarrow (S+S_3)$$

$$H(S) = \frac{100}{(S+2)(S+5)} \leftrightarrow$$

$$\frac{100}{3}\left\{\frac{1}{S+2} - \frac{1}{S+5}\right\} = H(S)$$

$$h(t) = \frac{100}{3}\left(e^{-2t} - e^{-5t}\right)$$

If $C(t) = k\left(e^{-2t} - e^{-500t}\right)$

$t = 1 \quad C(t) = ke^{-2t}$

17. $T(S) = \dfrac{100}{(S+2)(S+12)}$

Solution:

$p_1 = 2$

$$\frac{p_2}{p_1} = 6 > 5 \quad p_2 = 12$$

$$\approx \frac{100}{(S+2)(12)} \approx \frac{\frac{100}{12}}{(S+2)}$$

18. $\dfrac{C(S)}{R(S)} = \dfrac{288}{(S+2)(S+5)(S+12)(S+24)}$

Solution:

$p_1 = 2$

$p_2 = 5$

$p_3 = 12$

$p_4 = 24$

$$\frac{p_2}{p_1} = \frac{5}{2} < 5$$

$$\frac{p_3}{p_1} = \frac{12}{2} > 5$$

$$\frac{p_4}{p_1} = \frac{24}{2} > 5$$

$$\approx \frac{288}{12 \times 24(S+2)(S+5)}$$

$$= \frac{1}{(S+2)(S+5)}$$

19. $\dfrac{C(S)}{R(S)} = \dfrac{5}{(S+5)(S^2+S+1)}$

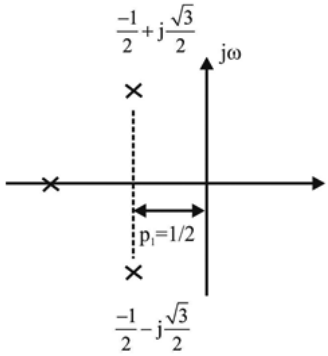


Fig. 5.77

$$\frac{p_2}{p_1} = 10 > 5$$

$$\approx \frac{5}{5(S^2+S+1)} = \frac{1}{(S^2+S+1)}$$

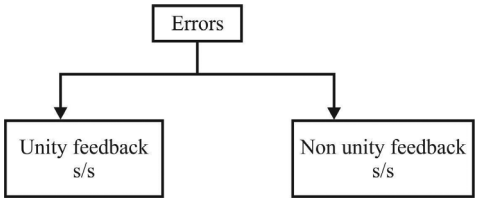


Fig. 5.78

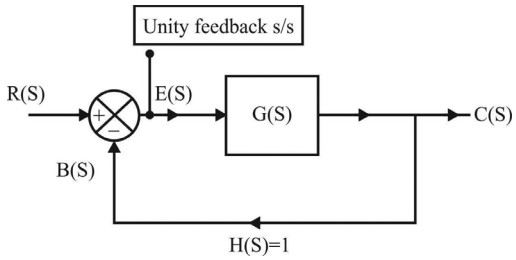


Fig. 5.79

$B(S) = C(S)$

$E(S) = Error\ \ Signal$

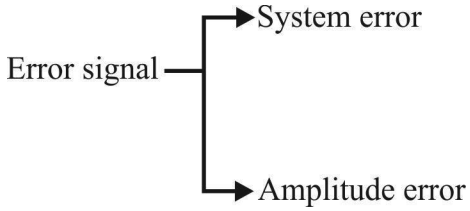


Fig. 5.80

System error:

1. System error is zero as $t \to t_s$
2. A system always minimizes system error

$r(t) = Au(t) \to e_s(t) = r(t) - c(t)$

$r(t) = A\delta(t) \to e_s(t) = \int_{-\infty}^{\infty} r(t)dt - \int_{-\infty}^{\infty} c(t)dt$

$r(t) = Atu(t) \to e_s(t) = \frac{dr(t)}{dt} - \frac{dc(t)}{dt}$

Amplitude error:

$e(t) = r(t) - c(t)$

It is always present at the output of summer

$r(t) = Au(t) \quad e(t) = Au(t) - c(t)$

$r(t) = A\delta(t) \quad e(t) = A\delta(t) - c(t)$

$r(t) = Atu(t) \quad e(t) = Atu(t) - c(t)$

Calculation of amplitude error:

1. $e(t) = r(t) - c(t) \to$ At the output of summer
2. $E(S) = R(S) - C(S) \to$ At the output of summer
3. Steady-state amplitude error:

$\lim_{t\to\infty} e(t) = \lim_{S\to 0} SE(S) \to$ All the poles of SE(S) must be strictly in L.H.P.

$\frac{C(S)}{R(S)} = \frac{G(S)}{1+G(S)} \qquad E(S) = R(S) - C(S)$

$E(S) = R(S) - \frac{G(S)R(S)}{1+G(S)} = R(S)\left\{1 - \frac{G(S)}{1+G(S)}\right\}$

$E(S) = \frac{R(S)}{1+G(S)}$

$\lim_{t\to\infty} e(t) = \lim_{S\to 0} \frac{SR(S)}{1+G(S)} \qquad \frac{S+R(S)}{\{1+G(S)\}}$

All poles must be strictly in LHP

other wise

Indeterminate Infinite

Calculation of steady-state amplitude error for different inputs:

1. $r(t) = Au(t) \leftrightarrow R(S) = \frac{A}{s}$

$e_{ss} = \lim_{S\to 0} \frac{A}{1+G(S)} = \frac{A}{1+\lim_{S\to 0} G(S)} = \frac{A}{1+k_p}$

k_p = Positional error constant or coefficient.

2. $r(t) = Atu(t) \leftrightarrow R(S) = \frac{A}{S^2}$

$e_{ss} = \lim_{S\to 0} \frac{A}{S(1+G(S))} = \lim_{S\to 0} \frac{A}{S+SG(S)}$

$= \lim_{S\to 0} \frac{A}{SG(S)} = \frac{A}{k_v}$

$k_v = \lim_{S\to 0} SG(S) \to$ Velocity error constant or coefficient

3. $r(t) = \frac{At^2}{2}u(t) \leftrightarrow R(S) = \frac{A}{s^3}$

$e_{ss} = \lim_{S\to 0} \frac{A}{S^2(1+G(S))} = \frac{A}{\lim_{S\to 0} S^2G(S)} = \frac{A}{k_a}$

k_a = Acceleration error constant.

Points To Remember

Input	Limitation	Steady-state error
Au(t)	**SE(S)** → all poles in LHP	$e_{ss} = \frac{A}{1+k_p} \rightarrow k_p = \lim_{s\to 0} G(S)$
Atu(t)	**SE(S)** → all poles in LHP	$e_{ss} = \frac{A}{k_v} \rightarrow k_v = \lim_{s\to 0} SG(S)$
$\frac{At^2}{2}u(t)$	**SE(S)** → all poles in LHP	$e_{ss} = \frac{A}{k_a} \rightarrow k_a = \lim_{S\to 0} S^2G(S)$

Table 5.2

Alternate expression for amplitude error:

e(t) = r(t) – c(t)

E(S) = R(S) – C(S) $\quad \frac{C(S)}{R(S)} = T(S)$

E(S) = R(S) – R(S) T(S)

E(S) = R(S) {1 – T(S)}

$\lim_{t\to\infty} e(t) = \lim_{S\to 0} SE(S) = \lim_{S\to 0} SR(S)\{1-T(S)\}$

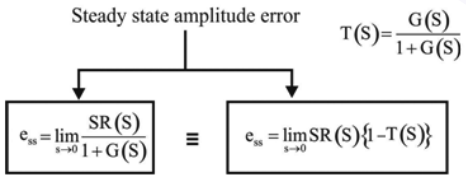


Fig. 5.81

Steady-state amplitude error for step input:

$r(t) = Au(t)$

$e_{ss} = \frac{A}{1+k_p}$

Case 1: $G(S) = \frac{N(S)}{S^0D(S)} \rightarrow Type\ '0'$

$k_p = \lim_{S\to 0} G(S) \neq 0$

$e_{ss} = \frac{A}{1+k_p}$

Case 2: $G(S) = \frac{N(S)}{SD(S)} \rightarrow Type\ '1'$

$k_p = \lim_{S\to 0} G(S) = \infty$

$e_{ss} = \frac{A}{1+k_p} = 0$

Case 3: $G(S) = \frac{N(S)}{S^2D(S)} \rightarrow Type\ '2'$

$k_p = \lim_{S\to 0} G(S) = \infty$

$e_{ss} = 0$

Steady-state amplitude error for ramp input:[r(t) = Atu(t)]

$e_{ss} = \frac{A}{k_v}$

Case 1: $G(S) = \frac{N(S)}{S^0D(S)} \rightarrow type\ '0'$

$k_v = \lim_{S\to 0} SG(S) = 0$

$e_{ss} = \frac{A}{k_v} = \infty$

Case 2: $G(S) = \frac{N(S)}{S^1D(S)} \rightarrow type\ '1'$

$k_v = \lim_{S\to 0} SG(S) \neq 0$

$e_{ss} = \frac{A}{k_v}$

Case 3: $G(S) = \frac{N(S)}{S^2D(S)} \rightarrow type\ '2'$

$$k_v = \lim_{S\to 0} SG(S) = \infty$$

$$e_{ss} = \frac{A}{k_v} = 0$$

Steady-state amplitude error for parabolic input:

$$r(t) = \frac{At^2}{2}u(t)$$

$$e_{ss} = \frac{A}{k_a}$$

$$G(S) \to Type\ 0 \to e_{ss} = \infty$$

$$G(S) \to Type\ 1 \to e_{ss} = \infty$$

$$G(S) \to Type\ 2 \to e_{ss} = finite$$

G(s) → Type 3 → e_{ss} = 0

Important Points

- As the type of system increases, the ability of the system to accept more no. of inputs increases.
- $r(t) = Au(t - T_0) \leftrightarrow R(S) = \frac{Ae^{-sT_0}}{S}$

$$e_{ss} = \lim_{S\to 0} \frac{SR(S)}{1+G(S)} = \lim_{S\to 0} \frac{S \times Ae^{-ST_0}}{S(1+G(S))}$$

$$= \lim_{S\to 0} \frac{A}{1+G(S)}$$

- The steady-state error is independent of the time shift in input.
- The steady-state error can also be undefined for a stable closed-loop system.

$$r(t) = e^{2t}u(t)$$

Let all poles 1 + G(S) one in LHP CLS is stable

$$e_{ss} = \lim_{s\to 0}\left\{\frac{SR(S)}{G(s)}\right\} = \lim_{s\to 0}\left\{\frac{S}{(S-2)(1+G(s))}\right\}$$

Input	Feedback	e_{ss}
Au(t)	Unity	$\frac{A}{1+k_p}$
Atu(t)	Unity	$\frac{A}{k_v}$
$\frac{At^2}{2}u(t)$	Unity	$\frac{A}{k_a}$

Type	Step	Ramp	Parabolic
0	$\frac{A}{1+k_p} = e_{ss}$	$e_{ss} = \infty$	$e_{ss} = \infty$
1	$e_{ss} = 0$	$e_{ss} = \frac{A}{k_v}$	$e_{ss} = \infty$
2	$e_{ss} = 0$	$e_{ss} = 0$	$e_{ss} = \frac{A}{k_a}$

- $r(t) = Au(t) + Btu(t) + Ct^2u(t)$

$$e_{ss} = \frac{A}{1+k_p} + \frac{B}{k_v} + \frac{2C}{k_a}$$

Error analysis of non-unity feedback system:

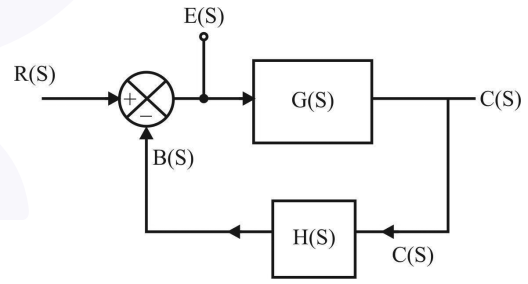


Fig. 5.82

Can be represented in diagram

1. R(S) – B(S) = E(S)
2. R(S) – C(S) = E_0(S)

It cannot be represented in a diagram.

Representation and calculation of actuating error:

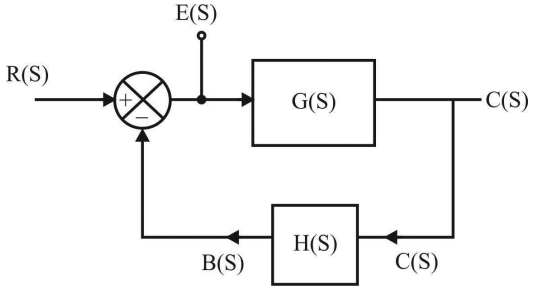


Fig. 5.83

Note: If the error is represented in the black diagram, then it is E(S)

E(S) = R(S) – B(S)

B(S) = C(S) H(S)

E(S) = R(S) – C(S) H(S)

$$E(S) = R(S) - \left\{\frac{R(S)G(S)}{1+G(S)H(S)}\right\}H(S)$$

$$E(S) = \frac{R(S)}{1+G(S)H(S)}$$

Steady-state error:

$$\lim_{t\to\infty} e(t) = \lim_{S\to 0}\left\{\frac{SR(S)}{1+G(S)H(S)}\right\}$$

All poles are strictly LHP.

Step input: $e_{ss} = \frac{A}{1+k_p}$ $\quad k_p = \lim_{S\to 0} G(S)H(S)$

Ramp input:

$$e_{ss} = \frac{A}{1+k_v} \qquad k_v = \lim_{S\to 0} SG(S)H(S)$$

Parabolic input:

$$e_{ss} = \frac{A}{1+k_a} \qquad k_a = \lim_{S\to 0} S^2G(S)H(S)$$

26. The O.L.T.F. of non-unity (–)ve feedback closed-loop system is given by

$$G(S)H(S) = \frac{K(1+S)}{S^2(S+20)}$$

The steady-state error for input $r(t) = 5 + 5t + 10t^2$, is $e_{ss} = 20$

Calculate K =?

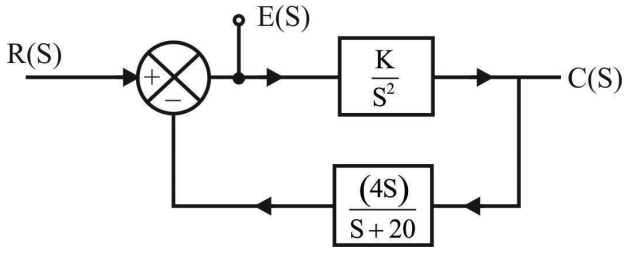


Fig. 5.84

Solution: $e_{ss} = \frac{5}{1+k_p} + \frac{5}{k_v} + \frac{20}{k_a} = 20$

$$G(S)H(S) = \frac{K(1+S)}{S^2(S+20)} \text{ Type II}$$

$$\frac{20}{k_a} = 20 \qquad k_a = \lim_{S\to 0} S^2G(S)H(S) = \frac{K}{20}$$

$$k_a = 1 \quad \frac{k}{20} = 1 \to k = 20$$

Concept of disturbance or noise signal as input:

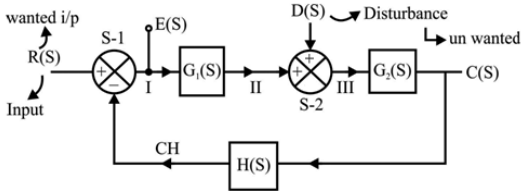


Fig. 5.85

Note: In general, noise or disturbance is additive in nature.

Calculation of E(S) = ?

(I) R – CH

(II) $G_1(R - CH)$

(III) $G_1(R - CH) + D$

$$C = \left[G_1(R - CH) + D\right]G_2$$

$$C = G_1G_2(R - CH) + DG_2$$

$$C = G_1G_2R - G_1G_2HC + DG_2$$

$$C\left[1 + G_1G_2H\right] = G_1G_2R + DG_2$$

$$C = \left\{\frac{G_1G_2}{1+G_1G_2H}\right\}_{\text{due to } R} R + \left\{\frac{G_2}{1+G_1G_2H}\right\}_{\text{due to } D} D$$

From (1)

$$E = R - CH$$

$$E = R - \left\{\frac{G_1G_2RH}{1+G_1G_2H}\right\} - \left\{\frac{G_2DH}{1+G_1G_2H}\right\}$$

$$E = R\left\{1 - \frac{G_1G_2H}{1+G_1G_2H}\right\} - \frac{G_2HD}{1+G_1G_2H}$$

$$E = \left\{\frac{R}{1+G_1G_2H}\right\} + \left\{\frac{-G_2HD}{1+G_1G_2H}\right\}$$

Steady-state error due to input and disturbance:

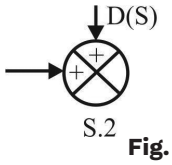
$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(S)$$

$$\lim_{t\to\infty} e(t) = \left(\lim_{s\to 0} \frac{SR(S)}{1+G_1G_2H}\right) + \left(\lim_{s\to 0} \frac{-SD(S)}{1+G_1G_2H}\right) \times G_2H$$

Steady-state error due to r(t) at summer 1

Steady-state error due to noise or disturbance at summer 1

Case 1:

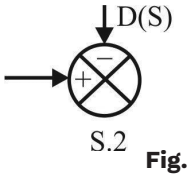


$H(S) \neq 1$

Fig. 5.86

$$e_{ss} = \lim_{S\to 0} \frac{SR(S)}{1+G_1G_2H} + \lim_{S\to 0}\left\{\frac{-SD(S)}{1+G_1G_2H}\right\} G_2H$$

Case 2:



$H(S) \neq 1$

Fig. 5.87

$$e_{ss} = \lim_{S\to 0} \frac{SR(S)}{1+G_1G_2H} + \lim_{S\to 0}\left\{\frac{+SD(S)}{1+G_1G_2H}\right\} G_2H$$

Steady-state error at S1 due to disturbance signal:

$$e_{ss} = \lim_{S\to 0} \frac{-SD(S)}{1+G_1G_2H} G_2H$$

Let

$$d(t) = u(t) \leftrightarrow D(S) = \frac{1}{s}$$

$H(S) = 1 \rightarrow$ *unity feedback*

$$e_{ss} = \lim_{s\to 0} \frac{-S \times \frac{1}{S} G_2}{1+G_1G_2} = \lim_{S\to 0} \frac{-G_2}{1+G_1G_2}$$

$$|e_{ss}| = \lim_{S\to 0} \frac{G_2}{1+G_1G_2}$$

$$|e_{ss}| = \lim_{S\to 0} \frac{1}{\frac{1}{G_2} + G_1}$$

#Magnitude of steady-state error due to disturbance is to be minimised:

G_1 should be as high as possible

G_2 should be as low as possible

Note:

1. In case of the presence of a disturbance signal (undesired), the error is considered at the output of the summer, where the desired input is being applied.
2. All the above results are based on the analysis of the standard block diagram.

30.

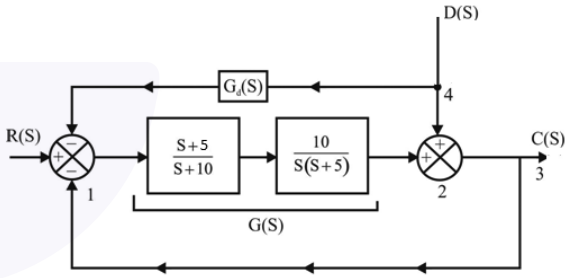


Fig. 5.88

The forward transfer function $G_d(S)$ is used to eliminate the effect of D(S) from the output. Calculate $G_d(S) = ?$

$$\left.\frac{C(S)}{D(S)}\right\}_{R(S)=0} = ?$$

Solution:

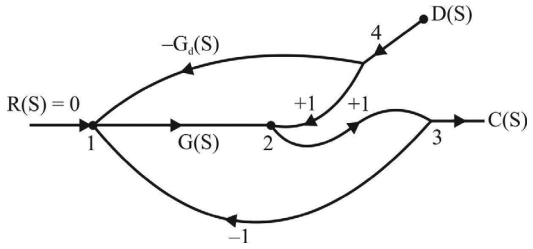


Fig. 5.89

$P_1 = +1$ $\quad \Delta_1 = 1$

$P_2 = -G_dG$ $\quad \Delta_2 = 1$

$$\left.\frac{C(S)}{D(S)}\right|_{R(S)=0} = \frac{1-G_dG}{1+G}$$

$$C(S)=\frac{(1-G_dG)}{(1+G)}D(S)\to 0$$

$$G_d=\frac{1}{G}\qquad G=\frac{10}{S(S+10)}$$

$$G_d=\frac{S(S+10)}{10}$$

Summary

Error calculation

1. Unity feedback:

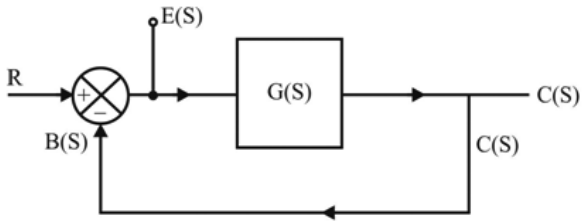


Fig. 5.90

Actuating error: $e_a(t)=r(t)-b(t)$

Output error: $e_0(t)=r(t)-C(t)$

$b(t)=C(t)$

$e_a(t)=e_0(t)$

$$e_{ss}=\lim_{S\to 0}\frac{SR(S)}{1+G(S)}$$

2. Non-unity feedback:

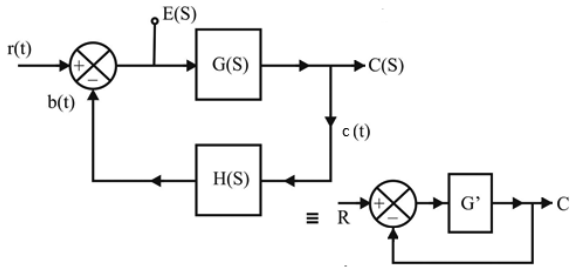


Fig. 5.91

1. $e_a(t)=r(t)-b(t)$

$$e_{ss}=\lim_{S\to 0}\frac{SR(S)}{1+G(S)H(S)}$$

2. $e_0(t)=r(t)-C(t)$

$$e_{ss}=\lim_{S\to 0}\frac{SR(S)\left(1+G(S)H(S)-G(S)\right)}{1+G(S)H(S)}$$

$$e_{ss}=\lim_{S\to 0}\frac{SR(S)}{1+G'(S)}$$

33. The steady-state error in a system is defined as.

$$\left\{\frac{dr(t)}{dt}-\frac{dc(t)}{dt}\right\}_{t\to\infty}\to \textit{slope error}$$

Where r is input and C is system output. Calculate steady-state error for input $t^3u(t)$ to a unity feedback system with $G(S)=\frac{100(S+1)(S+2)}{S^2(S+3)(S+10)}$.

Solution:

$$e(t)=\frac{dr(t)}{dt}-\frac{dc(t)}{dt}\to E(S)=S\left[R(S)-C(S)\right]$$

$$\lim_{t\to\infty}e(t)=\lim_{S\to 0}SE(S)\frac{C(S)}{R(S)}=\frac{G(S)}{1+G(S)}$$

$$\lim_{t\to\infty}e(t)=\lim_{S\to 0}\frac{S^2R(S)}{1+G(S)}\qquad t^nu(t)\leftrightarrow\frac{n!}{S^{n+1}}$$

$$\lim_{t\to\infty}e(t)=\lim_{S\to 0}\frac{S^2R(S)}{1+G(S)}\qquad t^3u(t)\leftrightarrow\frac{3!}{S^4}$$

$$=\lim_{S\to 0}\frac{6}{S^2\left(1+G(S)\right)}=\frac{9}{10}=0.9$$

Note: Let error is e'(t) = r'(t) – c'(t) or

$$\frac{de(t)}{dt}=\frac{dr(t)}{dt}-\frac{dc(t)}{dt}$$

$$e_{ss}=\lim_{S\to 0}\frac{S^2R(S)}{1+G(S)}\to \textit{Finite}:$$

$$\lim_{S\to 0}\frac{S^2\times 10}{S^2\left(1+G(S)\right)}=\lim_{S\to 0}\frac{10}{1+G(S)}$$

$$=\frac{10}{1+6k}=\frac{1}{6000}$$

n = 0 k = 9999.833

$k_p=\lim_{S\to 0}G(S)H(S)\to \textit{Finite}$

$k_v=\lim_{S\to 0}SG(S)H(S)\to 0$

$k_a=\lim_{S\to 0}S^2G(S)H(S)\to 0$

Cascading of 2ndOrder Underdamped and First-Order Systems

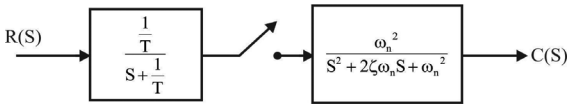


Fig. 5.92

$0 < \zeta < 1$

STD 2nd order TF t_s

$$\frac{C(S)}{R(S)} = \frac{\frac{1}{T}\omega_n^2}{\left(S+\frac{1}{T}\right)\left(S^2+2\zeta\omega_n S+\omega_n^2\right)} \rightarrow 3^{rd} \text{ order s/s.}$$

Case 1:

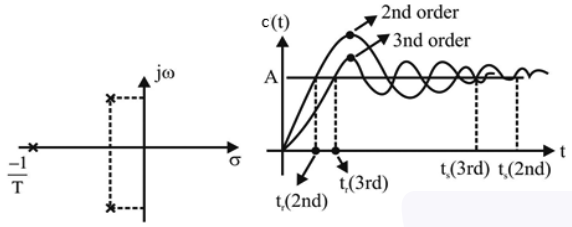


Fig. 5.93

Case 2:

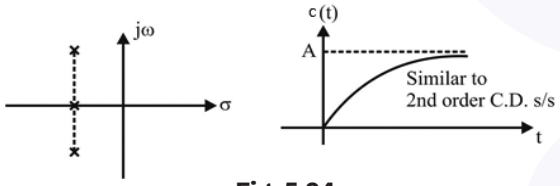


Fig. 5.94

Case 3:

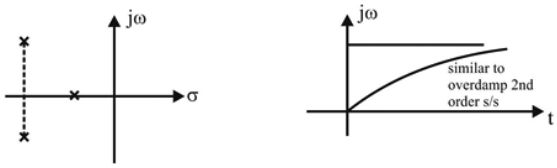


Fig. 5.95

Important Points

$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^2+2\zeta\omega_n S+\omega_n^2}$$

$$\frac{C(S)}{R(S)} = \frac{kN(S)}{\left(S^2+2\zeta\omega_n S+\omega_n^2\right)}$$

$$\frac{C(S)}{R(S)} = \frac{k\omega_n^2}{S^3+\alpha S^2+\beta S+r}$$

$$= \frac{k\omega_n^2}{(S+p)\left(S^2+2\zeta\omega_n S+\omega_n^2\right)}$$

$\rightarrow$ *Formula Invalid*

Test Your Knowledge

1. The O.L.T.F of a system is given as $G(S)H(S) = \frac{k}{S(S+1)(S+3)}$. If the corresponding closed-loop system is a unity (–)ve feedback system. Calculate gain "k" such that settling time is maintained at 12 secs for underdamped nature.

Solution: $k = \frac{70}{27}$

2. The forward gain of unity (–)ve feedback system is given as

$$G(S) = \frac{k(S+2)}{S^3+\beta S^2+4S+1}$$

Calculate the value of k and β such that Underdamped system has $\zeta = 0.2$ *and* $\omega_n = 3rad/\sec$

Solution: k = 7

$\beta = 2.87$

Exercise Problems

1. Consider the following closed-loop control system.

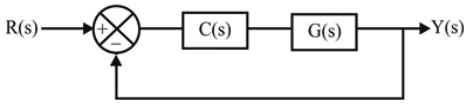


Fig. 5.96

$G(S)=\frac{1}{s(s+1)}$ and $C(s)=K\frac{s+1}{s+3}$. If the steady-state error for a unit ramp input is 0.1, then the value of K is _____.

Solution: 30

2. The open-loop transfer function of a system is $G(s)=\frac{(s+1)}{s^p(s+2)(s+3)}$, where p is an integer and is connected to unity (-) ve feedback configuration as shown in the figure.

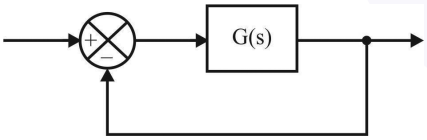


Fig. 5.97

Given that the steady-state error is zero for a unit step input and is 6 for unit ramp input, the value of the parameter p is _____.

Solution: 1

3. Consider a causal second-order system with the transfer function $G(s)=\frac{1}{1+2s+s^2}$

Let C(s) be the output with the unit step $R(s)=\frac{1}{s}$ as an input. The time taken by the system output c(t) to reach 94% of its steady-state value $\lim_{t\to\infty} c(t)$, rounded off to two decimal places, is

(A) 3.89 (B) 5.25

(C) 4.50 (D) 2.81

Solution: (C)

4. The open-loop transfer function of a unity (-)ve feedback control system is given by $G(s)=\frac{K}{s(s+2)}$

For the peak overshoot of the closed-loop system, to a unit step input to be 10%, the value of K is _____. **[GATE Set-01]**

Solution: 2.872

5. The position control of a D.C servomotor is given in the figure. The values of the parameters are

$K_T=1N-m/A, R_a=1\Omega, L_a=0.1H, J$
$=5\,kg-m^2, B=1N-m/(rad/\sec)$

and $K_b=1V/(rad/\sec)$.

The steady-state position response (in radians) due to unit impulse disturbance torque T_d is _____. **[GATE Set-03]**

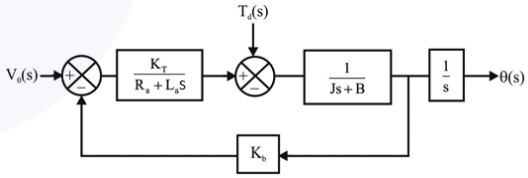


Fig. 5.98

Solution: -0.5

6. The output of a standard second-order system for a unit step input is given as

$y(t)=1-\frac{2}{\sqrt{3}}e^{-t}\cos\left(\sqrt{3}t-\frac{\pi}{6}\right)$

The transfer function of the system is **[GATE Set-02]**

(A) $\frac{2}{(s+2)(s+\sqrt{3})}$ (B) $\frac{1}{s^2+2s+1}$

(C) $\frac{3}{s^2+2s+3}$ (D) $\frac{4}{s^2+2s+4}$

Solution: (D)

7. For a unity feedback control system shown in the figure, the open-loop transfer function G(s) is given as $G(s)=\frac{2}{s(s+1)}$. The steady-state error e_{ss} due to a unit step input is

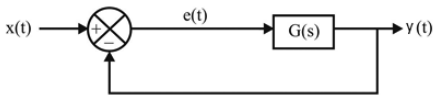


Fig. 5.99

(A) 0 (B) 0.5

(C) 1.0 (D) ∞

Solution: (A)

8. The damping ratio of the series RLC circuit can be expressed as **[GATE Set-01]**

(A) $\frac{R^2C}{2L}$ (B) $\frac{2L}{R^2C}$

(C) $\frac{R}{2}\sqrt{\frac{C}{L}}$ (D) $\frac{2}{R}\sqrt{\frac{L}{C}}$

Solution: (C)

9. A negative unity feedback system has an open-loop transfer function

$$G(s)=\frac{K}{s(s+10)}$$

The gain K of the system to have a damping ratio of 0.25 is _____. **[GATE Set-02]**

Solution: 400

10. The natural frequency of an undamped second-order system is 40 rad/s. If the system is damped with a damping ratio 0.3, the damped natural frequency in rad/sec is _____. **[GATE Set-02]**

Solution: 38.16

Exercise Problems Solution

1. Solution: 30

Given: Consider the following closed-loop

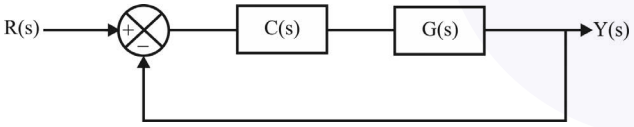


Fig. 5.100

Where $G(s)=\frac{1}{s(s+1)}$

$$C(s)=\frac{K(s+1)}{(s+3)}$$

Open-loop transfer function is given by,

OLTF = G(s)C(s)

$$G'(s)=\frac{1}{s(s+1)}\frac{K(s+1)}{(s+3)}=\frac{K}{s(s+3)}$$

Given: For ramp input, $e_{ss}=0.1$

e_{ss} for unit ramp input = $\frac{1}{K}$

Where $K_v \simeq$ *Velocity error* constant

$$K_v=\lim_{s\to 0} s[OLTF]$$

$$K_v=\lim_{s\to 0} sG'(s)=\lim_{s\to 0} s\frac{K}{s(s+3)}=\frac{K}{3}$$

Steady-state error is given by,

$$e_{ss}=\frac{1}{K_v}$$

$$e_{ss}=\frac{1}{K/3}=\frac{3}{K}$$

So, $\frac{3}{K}=0.1$

$K=30$

Hence, the correct answer is 30.

2. Solution: 1

Given: $G(s)=\frac{(s+1)}{s^p(s+2)(s+3)}$

$$e_{ss}\big|_{unit\ step\ input}=0$$

$$e_{ss}\big|_{unit\ ramp\ input}=6$$

For step input:

Steady-state error is given by,

$$e_{ss}=\frac{1}{1+K_p}$$

Where K_p = position error coefficient

$$\frac{1}{1+K_p}=0\Rightarrow K_p=\infty$$

The position error coefficient is given by,

$$K_p = \lim_{s\to 0} G(s)$$

$$K_p = \lim_{s\to 0} \frac{(s+1)}{(s+3)(s+2)} = \infty$$

$$K_p = \lim_{s\to 0} \frac{1}{6\times s^p} = \infty \Rightarrow p > 0$$

For ramp input:

Steady-state error is given by,

$$e_{ss} = \frac{1}{K_v}$$

Where K_v = Velocity error coefficient.

$$\frac{1}{K_v} = 6 \Rightarrow K_v = \frac{1}{6}$$

The velocity error coefficient is given by,

$$K_v = \lim_{s\to 0} sG(s) = \frac{1}{6}$$

The velocity error coefficient is given by,

$$K_v = \lim_{s\to 0} sG(s) = \frac{1}{6}$$

The velocity error coefficient is given by,

$$K_v = \lim_{s\to 0} sG(s) = \frac{1}{6}$$

$$\lim_{s\to 0} s \times \frac{(s+1)}{s^p(s+3)(s+2)} = \frac{1}{6}$$

$$\lim_{s\to 0} \frac{1}{6\times s^{p-1}} = \frac{1}{6}$$

$$s^{p-1} = s^0 \Rightarrow p = 1$$

Hence, the value of the parameter p is 1.

3. Solution: (C)

Given: $G(s) = \frac{1}{1+2s+s^2}$

$$R(s) = \frac{1}{s}$$

Method 1

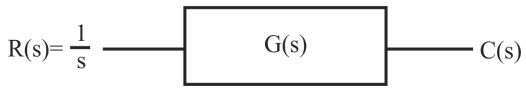


Fig. 5.101

$$c(s) = \frac{1}{s(s+1)^2}$$

By partial fraction

$$c(s) = \frac{A}{s} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2}$$

$$A = 1, B = -1, C = -1$$

$$c(s) = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

$$c(t) = 1 - e^{-t} + te^{-t}$$

$$0.94 = 1 - e^{-t} - te^{-t}$$

The above expression is satisfied at t = 4.50.

So, t = 4.50s

Hence, the correct option is (C).

4. Solution: 2.872

Given: $G(s) = \frac{K}{s(s+2)}$

$\%MPO = 10\%$

The characteristics equation is given by,

$$1 + G(s) = 0$$

$$1 + \frac{K}{s(s+2)} = 0$$

$$s^2 + 2s + K = 0 \qquad \ldots(i)$$

Standard second-order characteristic equation is given by,

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0 \qquad \ldots(ii)$$

From equations (i) and (ii),

$$\omega_n^2 = K \qquad \Rightarrow \omega_n = \sqrt{K}\ rad/\sec$$

$$2\xi\omega_n = 2$$

$$\xi = \frac{1}{\omega_n} = \frac{1}{\sqrt{K}} \Rightarrow K = \frac{1}{\xi^2} \qquad \ldots(iii)$$

Peak overshoot is given by,

$$MPO = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

$$0.1 = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

$$\frac{-\xi\pi}{\sqrt{1-\xi^2}} = \ln(0.1)$$

$$\frac{\xi\pi}{\sqrt{1-\xi^2}} = 2.3$$

$\xi = 0.59$

From equation (i),

$$K = \frac{1}{\xi^2} = \frac{1}{(0.59)^2} = 2.872$$

Hence, the value of K is 2.872.

5. Solution: -0.5

Given: $K_T = 1N-m/A,\ R_a = 1\Omega,$

$L_a = 0.1H, J = 5kg-m^2,$

$B = 1N-m/(rad/\sec), K_b = 1V/(rad/\sec)$

Method: Block diagram

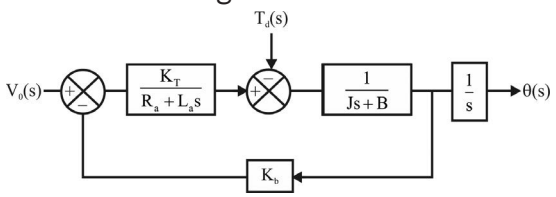


Fig. 5.102

For response due to disturbance, we consider $V_a(s) = 0$

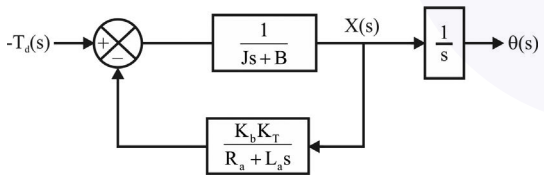


Fig. 5.103

$$\frac{X(s)}{-T_d(s)} = \frac{\frac{1}{Js+B}}{1+\frac{1}{(Js+B)}\times\frac{K_bK_T}{(R_a+L_as)}}$$

$$\frac{X(s)}{-T_d(s)} = \frac{R_a+L_as}{(Js+B)(R_a+L_as)+K_bK_T}$$

$$\frac{\theta(s)}{-T_d(s)} = \frac{\theta(s)}{X(s)}\times\frac{X(s)}{-T_d(s)}$$

$$\frac{\theta(s)}{-T_d(s)} = \frac{1}{s}\times\frac{X(s)}{-T_d(s)}$$

$$\frac{\theta(s)}{-T_d(s)} = \frac{R_a+L_as}{s\left[(Js+B)(R_a+L_as)+K_bK_T\right]}$$

For unit impulse disturbance torque,

$T_d(s) = 1$

$$\theta(s) = \frac{-1\times(R_a+L_as)}{s\left[(Js+B)(R_a+L_as)+K_bK_T\right]}$$

Steady-state response can be calculated using the final value theorem.

Applying the final value theorem,

$$\theta_{ss} = \lim_{s\to 0} s\,\theta(s)$$

$$\theta_{ss} = \lim_{s\to 0} s\frac{-s(R_a+L_as)}{s\left[(Js+B)(R_a+L_as)+K_bK_T\right]}$$

$$\theta_{ss} = \frac{-R_a}{BR_a+K_bK_T} = \frac{-1}{1+1} = -0.5$$

Hence, the steady-state position response is – 0.5.

6. Solution: (D)

Given: $y(t) = 1-\frac{2}{\sqrt{3}}e^{-t}\cos\left(\sqrt{3}t-\frac{\pi}{6}\right)$ ...(i)

Step response of the standard second-order system for unit step input is given by,

$$c(t) = 1-\frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin(\omega_d t+\theta)$$ ...(ii)

Where $\omega_d = \omega_n\sqrt{1-\xi^2}$,

$$\theta = \cos^{-1}\xi = \tan^{-1}\left[\frac{\sqrt{1-\xi^2}}{\xi}\right]$$

Comparing equations (i) and (ii),

$\xi\omega_n = 1$ ...(iii)

$$\frac{1}{\sqrt{1-\xi^2}} = \frac{2}{\sqrt{3}}$$ ...(iv)

$\omega_d = \omega_n\sqrt{1-\xi^2} = \sqrt{3}rad/\sec$ ...(v)

From equations (iii), (iv) and (v),

$\xi = \frac{1}{2}$

$\omega_n = 2\,rad/\sec$

The transfer function of second-order system is given by,

$$T(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} = \frac{4}{s^2+2s+4}$$

Hence, the correct option is (D).

7. Solution: (A)

The given block diagram is shown below,

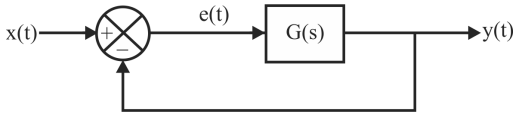


Fig. 5.104

$$G(s)=\frac{2}{s(s+1)} \qquad \text{[type-1 system]}$$

For step input (position input),steady-state error is given by,

$$e_{ss}=\frac{1}{1+K_p} \qquad ...(i)$$

Where K_p = Position error coefficient.

The position error coefficient is given by,

$$K_p=\lim_{s\to 0}G(s)=\lim_{s\to 0}\frac{2}{s(s+1)}=\infty$$

From equation (i),

$$e_{ss}=\frac{1}{1+\infty}=0$$

Hence, the correct option is (A).

8. Solution: (C)

Series RLC circuit in the transform domain is shown in the figure.

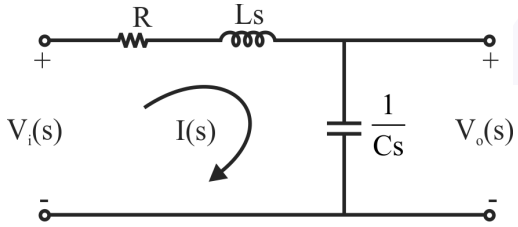


Fig. 5.105

$$V_0(s)=\frac{\frac{V_i(s)}{Cs}}{\left(R+Ls+\frac{1}{Cs}\right)} \qquad [By\ VDR]$$

$$\frac{V_0(s)}{V_i(s)}=\frac{1}{Cs\left(R+Ls+\frac{1}{Cs}\right)}=\frac{\frac{1}{LC}}{s^2+\frac{R}{L}s+\frac{1}{LC}} \qquad ...(i)$$

The transfer function of the second-order system is given by,

$$T(s)=\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \qquad ...(ii)$$

Comparing equations s (i) and (ii),

$$\omega_n^2=\frac{1}{LC}\Rightarrow\omega_n=\sqrt{\frac{1}{LC}}$$

$$2\xi\omega_n=\frac{R}{L}$$

$$\xi=\frac{R}{2L}\sqrt{LC}=\frac{R}{2}\sqrt{\frac{C}{L}}$$

Hence, the correct option is (C).

9. Solution: 400

Given: $G(s)=\frac{K}{s^2+10s},\xi=0.25$

The closed-loop transfer function for negative unity feedback is given by,

$$T(s)=\frac{G(s)}{1+G(s)}=\frac{\frac{K}{s^2+10s}}{1+\frac{K}{s^2+10s}}$$

$$T(s)=\frac{K}{s^2+10s+K} \qquad ...(i)$$

The transfer function of the second-order system is given by,

$$T(s)=\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \qquad ...(ii)$$

Comparing equations s (i) and (ii),

$$K=\omega_n^2\Rightarrow\omega_n=\sqrt{K}\,rad/\sec$$

$$2\xi\omega_n=10$$

$$\xi=\frac{10}{2\sqrt{K}}=0.25$$

$$\sqrt{K}=\frac{10}{0.5}=20$$

$$K=(20)^2=400$$

Hence, he gain K for the system is 400.

10. Solution: 38.16

Given: $\omega_n=40rad/\sec\ and\ \xi=0.3$

The damped natural frequency is given by,

$$\omega_d=\omega_n\sqrt{1-\xi^2}$$

$$\omega_d=40\sqrt{1-(0.3)^2}=38.16\,rad/\sec$$

Hence, the damped natural is 38.16. rad/sec.